

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-a58202b3689e77f52.jsonl`
- SHA-256: `890239e642b10c42a48460caee3bbcff97140d949755e012f54781d68168d7f1`
- Source modified: `2026-06-19T15:49:35+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_ac0cbb29-6bf`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-19T15:46:34.348Z)

You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements that CONTRADICT the current code or data.

HARD RULES:

- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured, say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.

Working dir is the repo root; use Read/Grep/Glob freely.

YOUR SLICE ? corpus:

GOLDEN-CORPUS SIZE + registry staleness. The corpus is now n=15 (output/human_lovo_manifest.json has 15 entries: adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles, mahadevpura_2, GX010128, mahadevpura_1, GX030094, Badminton_BXH_2, GX010141, GX010137, Boxhill_Doubles, GX020094). Find every doc stating a now-wrong CURRENT size (n=6/9/11/12/13) or a stale in-progress/pending-ingest status for clips that are in fact ingested. Files: docs/GOLDEN_VIDEOS.md (its registry table header says n=9 and lists clips ingestion-in-progress that are now ingested), docs/QUALITY_ITERATIONS.md (the "Golden corpus = the 6 labelled matches" line), docs/RALLY_DETECTION_GAP_CLOSURE.md (sections 3/6/7), docs/RALLY_QUALITY_RESEARCH.md, docs/GOLDEN_SET_*.md. Verify against output/human_lovo_manifest.json. IMPORTANT: leave clearly-HISTORICAL notes (at n=11 filed, changelog entries) alone ? only fix claims about the CURRENT/present corpus size or ingest status.

Return EVERY code-contradicting staleness item you can verify in your slice. Exhaustive within the slice; facts only.

2. assistant (2026-06-19T15:46:37.883Z)

I'll audit the corpus-related docs for staleness against the current golden corpus state. Let me start by verifying the ground truth.

3. user (2026-06-19T15:46:40.506Z)

count: 15

```
[
  {
    "name": "adarsh_avi_singles",
```

```

    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and
Avi.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv"
  },
  {
    "name": "largetest_doubles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv",
    "fps": 29.97,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv"
  },
  {
    "name": "gbaaddy",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/gBaaddy20May2026.rallies.csv"
  },
  {
    "name": "testlarge_short",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv",
    "fps": 29.97,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/TestLargeVideo.rallies.csv"
  },
  {
    "name": "mahadevpura_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv",
    "fps": 29.97,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/MahadevpuraSingles.rallies.csv"
  },
  {
    "name": "mahadevpura_2",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_2.rallies.csv",
    "trajectory": "C:/Users/avi

```

4. user (2026-06-19T15:46:42.683Z)

Your configuration specifies to merge with the ref 'refs/heads/fix/gemini-loud-chunk-failures' from the remote, but no such ref was fetched.

5. assistant (2026-06-19T15:46:45.473Z)

The manifest has 15 entries, confirming n=15. The git pull failed due to a branch config issue, but that's not blocking my read-only audit. Let me get the full list of names and then read the target docs.

6. user (2026-06-19T15:46:47.182Z)

```
['adarsh_avi_singles', 'kushagra_singles', 'largetest_doubles', 'gbaaddy', 'testlarge_short', 'mahadevpura_singles', 'mahadevpura_2', 'GX010128', 'mahadevpura_1', 'GX030094', 'Badminton_BXH_2', 'GX010141', 'GX010137', 'Boxhill_Doubles', 'GX020094']
```

7. user (2026-06-19T15:46:47.398Z)

```
1      # Golden-labeled video registry
2
3      **Purpose.** The single source of truth for every **golden** (human rally-boundary-
4      labeled) video ? what
5      it is, where the source proxy lives, how many rallies, and whether it's ingested into
6      the eval corpus.
7      Use this to **isolate the golden videos into a dedicated location** and to keep the
8      growing golden set
9      auditable. Updated whenever a new video is labeled + ingested.
10
11     > **This is the GROWING GOLDEN-finalized set** in the dual-eval-set discipline (see
12     > [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md), the 2026-06-16 capstone + `eval-dual-
13     set-regression`):
14     > a **RAW eval set** (currently-bad videos) + this **golden-finalized eval set**
15     (promoted when at-par) +
16     > a separate **golden TRAINING set**. Every quality launch must not regress on either
17     eval set.
18     > Privacy-safe, video-free *shipping* of these =
19     [GOLDEN_REGRESSION_FIXTURES.md](GOLDEN_REGRESSION_FIXTURES.md).
20
21     ## Contracts / conventions
22     - **Labels:** rally-annotator CSV
23     (`rally_number,start_time,end_time,ending_reason,sport,shots_count`,
24     decimal seconds) ? ingested by `backend/eval/gt_loader.py` (reads
25     `start_time`/`end_time`).
26     - **Alignment is mandatory before trusting any score.** The label timebase MUST match
27     the windowed proxy
28     (same fps + frame count). Verify with `ffprobe` (the `largetest_doubles` near-miss is
29     why) ? ideally
30     annotate the *exact* proxy the detector ran on (then `video_id` = MD5 matches and
31     alignment is free).
32     - **`video_id`** = MD5 of the proxy (`backend/utils/hashing.py::compute_video_id`); it
33     keys the WASB
34     trajectory cache, the Gemini reference rows, and the golden features.
35     - **Ingestion** = copy the trajectory + labels to durable paths, write
36     `output/<name>_golden_features.json`
37     (`gts` + candidates), append a row to `output/human_lovo_manifest.json`.
38     (`scratch/at_par.py` scores a
39     single video vs Gemini + golden; `backend/eval/rally_seg_eval.py` runs the whole
40     corpus.)
41     - **Guardrails:** owned footage only; **minors excluded** ; labels never derived from a
42     paid LLM (Gemini is
43     inference/reference only). Commercial-clean.
44
45     ## Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 9)
46
47     | # | name | sport | rallies | fps | source proxy | video_id | labeled |
48     |---|---|---|---|---|---|---|---|
49     | 1 | adarsh_avi_singles | badminton (S) | 96 | 59.94 | (original 6) | ? | pre-06-16 |
50     | 2 | kushagra_singles | badminton (S) | 32 | 59.94 | (original 6) | ? | pre-06-16 |
51     | 3 | largetest_doubles | badminton (D) | 49 | 29.97 | (original 6) | ? | pre-06-16 |
```

```

35      | 4 | gbaaddy | badminton | 48 | 59.94 | (original 6) | ? | pre-06-16 |
36      | 5 | testlarge_short | badminton | 6 | 29.97 | (original 6) | ? | pre-06-16 |
37      | 6 | mahadevpura_singles | badminton (S) | 31 | 29.97 | (original 6) | ? | pre-06-16 |
38      | 7 | mahadevpura_2 | badminton (multicourt) | 40 | 59.94 | `~/Roora Request - June
15/[Done] Video 3 ~/mahadevpura-2_proxy.mp4` | `038e0e0915a2` | 2026-06-16 |
39      | 8 | GX010128 | badminton (multicourt) | 52 | 59.94 | `~/Golden Label Videos Request -
June 15/[Done] Video 13 ~/GX010128.MP4` | `db84f3e65318` | 2026-06-16 |
40      | 9 | mahadevpura_1 | badminton (S, continuous) | 10 | 59.94 | `~/[Done] Video 14 -
Badminton/mahadevpura-1.MP4` | `38bef75d78cb` | 2026-06-17 |
41
42      ## Labeled, ingestion in progress (WASB trajectory extracting)
43
44      | name | sport | rallies | fps | source proxy | status |
45      |---|---|---|---|---|---|
46      | GX030094 | badminton | 41 | **29.97** | `~/Golden Label ? /[Done] Video 6 -
Badminton/GX030094_proxy.mp4` | WASB running ? ingest next |
47      | Badminton BXH 2 | badminton (multicourt) | 44 | 59.94 | `~/[Done] Video 10 - Badminton
Multicourt/Badminton BXH 2_proxy.mp4` | WASB running ? ingest next |
48
49      ## Source folder & not-yet-labeled
50      - **Annotation root:** `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos
Request - June 15\`
51      (+ the earlier `Roora Request - June 15\`). `[Done]` prefix in a dir name = golden-
labeled.
52      - **Labeled badminton dirs:** Video 3 (=mahadevpura-2), 6 (=GX030094), 10 (=BXH 2), 13
(=GX010128, a
53      re-label of #8), 14 (=mahadevpura-1). **Video 13 is a duplicate of #8** (already
ingested).
54      - **Not yet labeled / other sports:** Video 1, 4, 7, 9, 12 (badminton, pending); Video
2, 8 (table tennis);
55      Video 5, 11 (pickleball). Badminton is the current focus (alpha customers);
TT/pickleball labeled later.
56
57      ## How to isolate the golden videos
58      Each row's **source proxy** column is the file to copy into a dedicated golden-videos
location. The labels
59      live beside the proxy (`<name>.rallies.csv`) and a durable copy is in
`output/<name>.rallies.csv`; the
60      trajectory is in `Collect/Trajectories/` (path in the manifest). Keep this table in sync
on every new ingest.
61

```

8. assistant (2026-06-19T15:46:55.296Z)

GOLDEN_VIDEOS.md is clearly stale. The registry table header says n=9 but the manifest has 15 entries. Plus the GX030094 and Badminton_BXH_2 listed under "ingestion in progress" are now ingested. Let me verify the full manifest names match and check the other docs.

9. user (2026-06-19T15:46:56.934Z)

```

1 adarsh_avi_singles fps= 59.94
2 kushagra_singles fps= 59.94
3 largetest_doubles fps= 29.97
4 gbaaddy fps= 59.94
5 testlarge_short fps= 29.97
6 mahadevpura_singles fps= 29.97
7 mahadevpura_2 fps= 59.94
8 GX010128 fps= 59.94
9 mahadevpura_1 fps= 59.94
10 GX030094 fps= 29.97
11 Badminton_BXH_2 fps= 59.94
12 GX010141 fps= 59.94
13 GX010137 fps= 59.94
14 Boxhill_Doubles fps= 29.97
15 GX020094 fps= 29.97

```

10. user (2026-06-19T15:46:57.714Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-600 of 749 total (26506 tokens, cap 25000). Call Read with offset=601 limit=600 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      # Rally-detection quality iterations ? living log
2
3      **Purpose.** A running, reverse-chronological log of how we evolve rally-detection
quality: each
4      iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.
This is the
5      **live** continuation of the historical trail in
6      [`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the
archive policy
7      (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this
file stays at
8      `docs/` top level and keeps growing, then its entries graduate to the archive when they
conclude.
9
10     **Methodology (non-negotiable).** Every quality change rides the experiment harness
11     ([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default
OFF**;; they're
12     graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
happens **only
13     on measured lift** through the `run_regression.py` gate; **rollback = flip a config
flag, never
14     `git revert`.** Cues are designed in
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
15     in [EVALUATION.md](EVALUATION.md); the 2-layer model in
16     [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
17
18     **Golden corpus = the 6 labelled matches** in `output/human_lovo_manifest.json` (owner-
confirmed
19     2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, targetest_doubles,
gbaaddy,
20     testlarge_short, mahadevpura_singles`). The model trained on these is what we run
alongside the existing
21     detector on NEW footage for the side-by-side review.
22
23     ---
24
25     ## Baselines that drive strategy (IoU 0.5 vs human golden labels)
26
27     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
28     |---|---|---|
29     | Heuristic (velocity windowing) | 0.657 | 0.157 |
30     | Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
31     | Two-stage WASB+Gemini refine | 0.83 | ? |
32     | Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |
33
34     **Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the
commodity wrapper
35     to 0.66 ? that is the owned model's ship target.** Baselines tracked in
`eval_baselines/`. The owner's
36     observed false positive ? a player **holding/flicking** the shuttle between points
counted as a rally ?
37     is the precision bug the fusion gates target.
38
39     ---
40
41     ## Current track (live) ? newest first
42
43     ### P7 ? Long-rally (>5s) quality lens (eval) + reel-selection knob ? MEASURED, default-
```

```

OFF *(2026-06-18)*
44    **Hypothesis (owner).** Local OVER-FIRES, and its false positives are mostly SHORT
(motion blips,
45    background-court fragments on multicourt). Filtering the eval to **long rallies (>5s)**
? the
46    eye-catching highlights that carry the product ? should strip most short FPs and reveal
local's real
47    long-rally quality, answering the North-star "do we beat Gemini on the single-match
long-rally happy
48    path?". **Duration** (end?start), NOT shot-count, is the filter: golden `shots_count` is
blank and the
49    no-AI path reports it "Unknown", whereas duration comes from the start/end labels we
already trust.
50
51    **What shipped (both default-OFF, zero behaviour change at the defaults).**
52    - **Eval lens** ? `rally_seg_eval.py --min-duration <s>` filters BOTH predictions AND
golden GTs to
53    rallies longer than `s` *before* matching, consistently across the tIoU and R2
tolerance paths (a
54    FILTER, not a new metric). Plus `--stratify` / `--long-tau` (default 5s): a `{all vs
long} x {single
55    vs multicourt}` aggregate table. Court type = the committed `MULTICOURT_CLIPS` set
(the manifest is
56    gitignored, so the set is the auditable source), overridable by a manifest
`court_type` field; kept
57    in sync with [`GOLDEN_VIDEOS.md`](GOLDEN_VIDEOS.md).
58    - **Product knob** ? `stitching.min_rally_duration` (s, 0 = OFF) drops short rally pairs
before the
59    reel compile in `tools/generate_highlights.py` ? a cleaner reel-selection filter than
60    `min_shot_count` (which leans on the "Unknown" shot_count).
61
62    **Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m
backend.eval.rally_seg_eval --stratify`):**
63
64    | stratum | court | n | tolF1 | tolP | tolR | tIoU F1@0.5 | seg_ratio |
65    |---|---|---|---|---|---|---|---|
66    | all | single | 11 | 0.375 | 0.326 | 0.464 | 0.482 | 1.434 |
67    | all | multicourt | 4 | 0.294 | 0.238 | 0.393 | 0.430 | 1.601 |
68    | all | all | 15 | 0.353 | 0.302 | 0.445 | 0.468 | 1.478 |
69    | **long>5s** | single | 11 | 0.264 | 0.278 | 0.267 | 0.416 | **1.037** |
70    | **long>5s** | multicourt | 4 | 0.240 | 0.197 | 0.310 | 0.430 | **1.597** |
71    | **long>5s** | all | 15 | 0.257 | 0.256 | 0.279 | 0.420 | 1.186 |
72
73    Pooled counts: single **608 preds / 416 GTs** (only 36% of preds are >5s vs 50% of GTs ?
long pred:GT ?
74    **218:210 = 1.04**); multicourt **288 preds / 179 GTs** (55% of preds >5s ? long pred:GT
= **158:100 =
75    1.58**).
76
77    **Findings ? the lens turned a one-line hypothesis into a 3-part measured map (NUMBERS
INFORM):**
78    1. ? **Single court, count-level: thesis HOLDS.** Local's over-production is dominated
by SHORT windows
79    ? restricting to >5s collapses seg_ratio **1.43 ? 1.04** (preds ? GTs in count). The
short-blip FPs
80    are real and duration-filterable.
81    2. ?? **Multicourt: thesis FAILS.** Even among >5s rallies local over-fires **1.58x** ?
adjacent-court
82    games are genuine LONG rallies ? long *spurious* windows. **A duration filter does
NOT clean
83    multicourt; that needs court-localization** (Gate-B / court mask ? gap-closure Lever
C). So "FPs are
84    mostly short" is a *single-court* statement.
85    3. ?? **Strict tolerance precision does NOT jump** (tolP single 0.326 ? 0.278) even
where the count

```

86 normalizes, because the >5s pred set is contaminated by ****over-MERGED windows****
 (kushagra med 9.9s,
 87 mahadevpura_1's 63s blob, GX010141 14.8s) and cap=6 truncates genuinely-long rallies
 so the strict
 88 |?end| ? 1.5 s match fails (tIoU, forgiving, holds ~0.42; strict tol drops). The
 long-rally
 89 *strict-eval* happy path is gated by merge-control + the cap-vs-long tension, not
 just FP-stripping.
 90 4. ? ****Product knob is still a clean reel win.**** A reel doesn't need strict boundary
 tolerance ? an
 91 over-merged 15 s clip is a fine highlight, and dropping <5 s pairs removes the
 single-court short
 92 junk. Defaults OFF; owner reviews the numbers before any flip.
 93
 94 ****Gemini:**** not computable in-checkout ? the Boxhill gemini intervals live in the owner-
 side scorer's DB
 95 (not committed). The lens is metric-agnostic: feed Gemini's intervals through the same
 96 `filter\_by\_min\_duration` to get its long-rally row (owner-side one-liner). From gap-
 closure \$3, Gemini
 97 already fires fewer/longer windows with ~0 short FPs, so a long filter mostly affects
 LOCAL.
 98
 99 ****Lesson.**** A plausible product one-liner ("filter to long ? precision jumps") split
 into three measured
 100 truths once instrumented: single-court count ?, multicourt ? (needs a court signal),
 strict-precision ?
 101 (needs merge control). Default-OFF lens + knob; the owner decides any flip on the
 numbers. *(P7 in
 102 [`RALLY\_DETECTION\_GAP\_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md) \$7.)*
 103
 104 **### R1 ? MILESTONE LAUNCHED** (cap=6 + R4 flipped default-ON) ? re-verified at n=13
 (2026-06-18)
 105 The R1 milestone is ****shipped**** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are
 now ON.
 106 Full quality-impact table on both rulers, the over-seg guard, and the honest caveats are
 recorded in
 107 the ****[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)**** (per the Launch-
 Report convention).
 108
 109 ****Re-verified on the grown n=13 corpus**** (the 2 new Boxhill clips GX010137/GX010141) ?
 the milestone
 110 holds and the headline is ***stronger*** than at n=11:
 111
 112 | transition | n=11 (filed) | ****n=13 (launched)**** |
 113 |---|---|---|
 114 | baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | ****?+0.222, 12/0/1, p=0.0005**** |
 115 | cap12 ? cap6 tolF1 | ?+0.108, 10/0, p=0.002 | ****?+0.110, 12/0/1, p=0.0005**** |
 116 | cap6 ? +R4 tolF1 | ?+0.047, 8/0, p=0.008 | ?+0.040, ****9/1/3****, p=0.0215 |
 117 | net over-seg guard | PASS (mr 0.694?0.150) | ****PASS**** (mr 0.651?0.161, 0 regressions)
 |
 118
 119 Both new clips improve (GX010137 0.448?0.685, GX010141 0.333?0.431); tIoU ladder
 baseline 0.236 ?
 120 milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo\_resweep.py`,
 matches to 3 dp;
 121 LOVO picks cap=6 ****13/13**** on tIoU) and survived a 5-lens adversarial workflow (every
 L00 fold clears
 122 the bar; worst-fold drop GX010128 still ?+0.202, p=0.0010 ? no single clip drives it).
****Honest notes:****
 123 W1 cap is the dominant lever / R4 is the marginal +0.040 increment (1 loss = GX010137, a
 tolerance
 124 artifact); the gate is the NET rate-based guard (strict=False by design); GX010141 is
 the worst-behaved
 125 clip (future over-merge target). The nightly self-flags STALE post-flip ? re-baseline
 once with

```

126  `--update-baseline`. W2 / hard-cap / R5 stay default-OFF.
127
128  ### R5 ? blob-fallback for the continuously-active blob (default-OFF, stratum tool)
129  *(2026-06-17)*
130  The one clip the cap6+R4 milestone still fails is the **continuously-active blob**
131  (mahadevpura_1:
132  ~174 s of play with no inactivity gap to split on). The W1 gap-split finds no gap; R4
133  finds no slow
134  tail; the result is one ~66 s mega-window that matches nothing (**tolF1 0.000**, the
135  degenerate
136  0-match outcome [`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md)
137  $5 R5 names).
138
139  **R5** (`window_blob_fallback`, default-OFF) is the last resort: **after** the gap-split
140  and the R4
141  trim have each had their chance, any group still longer than `window_blob_factor` × the
142  cap is a
143  genuine blob ? midpoint-chop it to ? the cap, **log the forced cut**, and **flag every
144  resulting
145  window `forced_cut=True` (surfacing the clip as one that needs rally-STATE cues ? net-
146  crossings /
147  two-sided motion ? not a bigger cap). `_blob_fallback_chop` in `tracknet_runner`, wired
148  through
149  WindowingPreset / IndexingConfig / the served path / `rally_seg_eval --blob-fallback`.
150
151  **Why a factor gate, and why not just `window_hard_cap`: `hard_cap` chops **before**
152  R4 and fires on
153  **every** gap-less over-cap window ? it over-segments normal clips (mahadevpura_2 40?68).
154  R5 runs **after**
155  R4, and the magnitude gate (**default 4x**) leaves a normal long rally (~1?2× the cap)
156  alone, force-cutting
157  **only** the degenerate blob. That makes it surgical and do-no-harm.
158
159  **MEASURED (n=11, on the cap6+R4 milestone, R2 `tolF1`):**
160
161  | clip | milestone | + R5 | windows (milestone ? +R5) |
162  |---|---|---|---|
163  | **mahadevpura_1 (the blob)** | **0.000** | **0.238** | 2 (max 65.8 s) ? 32 (max 4.1
164  s), all `forced_cut` |
165  | mahadevpura_2 (also continuously-active) | 0.122 | 0.180 | 42 ? 100 |
166  | kushagra_singles | 0.078 | 0.110 | 45 ? 71 |
167  | Badminton_BXH_2 | 0.328 | 0.324 | 78 ? 92 (?0.004, noise) |
168  | *(the other 7 clips)* | ? | *unchanged* | no-op (no group exceeds 4× the cap) |
169
170  At factor=4, R5 touches **only** the 2 continuously-active clips + a noise-level BXH2
171  ?0.004 ? effectively
172  **do-no-harm**. It is a **STRATUM tool**: the win is the blob rescue (0.000?0.238,
173  removing the
174  mean-swinging 0-match pathology), not a corpus-mean lift (agg `tolF1` 0.323?0.354, sign
175  **3/3/5, n.s.** at
176  factor 4; the factor sweep is monotonic ? 6× is strictly do-no-harm, 3× over-chops).
177  **Default-OFF**; turn
178  it on selectively for continuously-active footage. The R1 net over-seg guard confirms
179  it: milestone?+R5 net
180  cost 0.423?0.289, **PASS**, no merge_rate regression. [numbers:
181  `scratch/r5_evidence.py`]
182
183  ### R1 ? milestone evidence (cap=6 + R4) + the net over-seg promotion guard
184  *(2026-06-17)*
185  The R1 milestone is the smallest safe local-windowing flip: **W1 bounded windowing cap=6
186  + the R4
187  rally-END inactivity trim, no W2** (the per-clip-best config, settled by R3/R4). This
188  entry (a) re-runs
189  the full ablation ladder baseline?milestone under the **R2** `tolF1` on **n=11** with
190  honest paired

```



```

167     sign-tests, and (b) ships the **net over-seg promotion guard** that unblocks the
milestone.
168
169     **The full ladder (n=11, R2 `tolF1`, asymmetric ?_start=2/?_end=1.5):**
170
171     | config | tolF1 (mean [95% CI]) | tIoU-F1 | mean \[?end\] |
172     |---|---|---|---|
173     | baseline (all-OFF) | 0.091 [0.026, 0.179] | 0.195 | 52.1 s |
174     | + W1 cap=12 | 0.168 [0.100, 0.241] | 0.357 | 6.5 s |
175     | + cap=6 (R3) | 0.276 [0.174, 0.380] | 0.407 | 5.7 s |
176     | ++ R4 inact_end=1.5 (milestone)** | **0.323 [0.210, 0.433]** | **0.440** | **3.8 s**
|
177
178     **Paired sign-tests (the promotion evidence):**
179     - **baseline ? milestone:** ?`tolF1` **+0.232** [+0.153, +0.304], **sign 10/0/1,
p=0.002** (the 1 tie is the
180     degenerate blob mahadevpura_1 ? both score 0.000; excluding it, 10/0). tIoU-F1 ref:
+0.245, 10/0, p=0.002.
181     - **cap=12 ? cap=6** (R3 was means-only ? now confirmed): ?`tolF1` **+0.108** [+0.070,
+0.154], **sign 10/0/1,
182     p=0.002**. **cap=6 beats cap=12, significantly.**
183     - **cap=6 ? +R4:** ?`tolF1` **+0.047** [+0.022, +0.072], **sign 8/0/3, p=0.008**
(reproduces the R3/R4 entry).
184
185     **The net over-seg promotion guard** (`tolerance_metrics.over_seg_guard`, surfaced in
186     `rally_seg_eval.compare()` / `--vs`). R2 §2.3.3's *strict* rule blocks promotion if the
candidate raises
187     `split_count` **or** `merge_count` on **any** video ? correct for an incremental cue,
but it **mis-fires on a
188     structural change** (mega-windows ? bounded): splitting a mega-window necessarily lifts
`split_count` from ~0,
189     and the raw merge **count** is non-monotonic. The milestone trips strict on 10/11
videos, **yet it strictly
190     reduces over-merge**:
191
192     - mean **`merge_rate`** (fraction of golden rallies swallowed in a merge) **0.694 ?
0.150**, dropping on **all
193     11** videos (? ?0.545, sign **0/11**, p=0.001). The "3 videos with a higher merge
*count*" (kushagra,
194     mahadevpura_1/2) all have *lower* merge *rate* ? one mega-window hiding 40 rallies is
`count=1, rate=1.0`;
195     bounding it into pieces that each glue 2 neighbours is `count=9, rate=0.5` (**fewer**
rallies hidden).
196
197     So the guard scores over-seg on the **normalized rates** (the comparable, recall-
meaningful quantity), as one
198     weighted cost per video ? `w_merge.merge_rate + w_split.split_rate`, **merges weighted
higher** (a merge hides a
199     rally = recall loss; a split only fragments one, recall preserved). It **PASSES** iff
(1) the corpus-mean net
200     cost does not rise **and** (2) no video's `merge_rate` regresses. Measured: net cost
**1.388 ? 0.423** (? ?0.966),
201     no per-video `merge_rate` regression ? **PASS at every weight w_merge?{1,2,3}** (the
raw-count basis flips on the
202     weight ? exactly why rates are the default). The strict verdict is always reported
alongside, and reverting to it
203     is one line (`use the verdict's `strict_passes``). [numbers: `scratch/r1_evidence.py`]
204
205     **R1 verdict:** the milestone (cap=6 + R4) clears the refined (net-guard) promotion bar
? a significant `tolF1`
206     win on the full n=11 corpus with **no honest over-seg regression**. The 2-line config
flip
207     (`indexing.max_window_duration: 6`, `window_inactivity_end: 1.5`,
`inactivity_rally_thresh: 0.20`,
208     `inactivity_min_density: 0.30`) is **prepped on its own branch and left for the owner's

```

```

product-go** ? it changes
209 shipped behaviour, so it is NOT flipped here. (W2 net-crossings stays default-OFF +
retained ? no GT-backed lift.)
210
211 ### R3 (cap re-scope) + R4 (rally-END inactivity trim) ? measured on n=11 golden under
R2 *(2026-06-17)*
212 The first quality levers scored on the **R2 task-faithful metric** (tolerance-match
`tolF1`, asymmetric
213 ?_start=2/?_end=1.5; see [R2_EVAL_METRIC_DESIGN.md](R2_EVAL_METRIC_DESIGN.md)), on the
**n=11** golden corpus
214 (now incl. 5 real ground-truth clips: mahadevpura_1/_2, GX010128, GX030094,
Badminton_BXH_2).
215
216 - **R3 ? re-scope the cap (12 ? 6 s).** The 12 s cap was too coarse. Sweep (W1, no R4):
`tolF1` cap=6 **0.273**
217 > cap=8 0.234 > cap=12 0.171. So R1's recommended cap is **6 s**, not 12. (Means-based
lead across the
218 sweep; confirm with the paired sign-test before locking the config value.)
219 - **R4 ? rally-END inactivity trim** (`window_inactivity_end`, **default-OFF**). The
measured gap is the
220 rally-*end* (|?end| ? |?start| on every ground-truth clip): after a rally the shuttle
keeps moving *slowly*
221 above `velocity_thresh` (picked up / walked), so the velocity window over-extends its
END. R4 trims that
222 post-rally drift tail back to the last frame whose trailing window stays dense with
**fast** motion
223 (velocity > `inactivity_rally_thresh`); only cuts (recall can't rise), START untouched
(already ?Gemini).
224 `_trim_inactive_tail` in `tracknet_runner`, wired through WindowingPreset /
IndexingConfig / served path /
225 `rally_seg_eval --inactivity-end`.
226
227 **MEASURED (n=11, on top of W1 cap=6; sweet spot ?_end-window=1.5 s,
rally_thresh=0.20, min_density=0.30):**
228
229 | metric | cap=6 | + R4 | ? |
230 |---|---|---|---|
231 | tolF1 | 0.276 | **0.323** | **+0.047** (95% CI [+0.022,+0.072], **sign-test 8/0,
p=0.008**) |
232 | mean \|?end\| | 5.71 s | **3.81 s** | ?1.9 s (toward Gemini's ~1.0) |
233 | mean split-count | 5.5 | 5.1 | net DOWN (only mahadevpura_2 +1; GX010128 4?2, BXH2
5?4) |
234
235 **R4 beats the hard-cap mechanism for ends:** hard-cap got |?end| 1.07 s but split
19.5 (heavy
236 over-production); R4 gets a *significant* tolF1 lift + tighter ends with splits **net-
down** ? it cuts at
237 the real inactivity point, not a blind midpoint. Cumulative tolF1: baseline 0.096 ?
cap12 0.171 ? cap6
238 0.276 ? +R4 **0.323**. Default-OFF; promote via config once R1's net-over-seg guard +
corpus growth land.
239
240 ### Nightly rally-quality regression tripwire ? the dual-eval-set discipline made
automatic *(2026-06-17)*
241 A scheduled nightly job re-scores the **golden corpus**
(`output/human_lovo_manifest.json`, n=11) under HEAD
242 and trips a wire if anything regresses vs a frozen baseline. It is a *tripwire*, **not**
a re-training run:
243 the WASB GPU trajectories are **CACHED** and only the windowing/scorer changes night to
night, so the re-score
244 is **CPU, seconds, no GPU/Gemini** (`scripts/nightly_regression.py` ?
`rally_seg_eval.evaluate`).
245
246 - **Two tracked configs**, both re-scored every night: **DEFAULT** (the SERVED
production windowing, all

```

247 milestone knobs OFF ? guards shipped behaviour) and **MILESTONE** (R3 cap=6 + R4
 inactivity trim ? guards the
 248 candidate **before** it's flipped ON). Frozen numbers (n=11, this commit): DEFAULT tolF1
0.091 / f1@0.5 0.194;
 249 MILESTONE tolF1 ***0.323*** / f1@0.5 0.440 (matches the R3/R4 row above ? cross-checks
 the harness).
 250 - ***What trips it:*** any gated aggregate drop beyond tolerance (``tol_f1`, tIoU`
``f1@0.5`/`f1@0.25``; ``merge_rate``
 251 rise) ***or*** any per-video **over-seg guard** rise (``split_count`/`merge_count`` may
 not increase on ANY video ?
 252 the R2 §2.3.3 guard). Boundary-error medians + seg_ratio are outlier-dominated ?
 reported, ***not*** gated. Exit
 253 1=regression, 4=stale (corpus/config changed ? re-snapshot), writes
``output/nightly_regression/<date>.md``.
 254 - ***Default-ADDITIVE ? it does NOT change the promotion gate.*** The baseline is tied to
 the LOCAL golden corpus
 255 (not committed) and is re-snapshotted (``--update-baseline``) after an intended
 improvement or when the corpus
 256 grows. Realises ``memory/eval-dual-set-regression`` (the growing GOLDEN-finalized set);
 the future **RAW eval**
 257 set (currently-bad videos) plugs in as a third config. Wired as a local Windows
 Scheduled Task
 258 (``scripts/nightly_regression.ps1`` + ``scripts/register_nightly_task.ps1``) ? a cloud
 agent can't run it because
 259 the cached trajectories live outside the repo.
 260
 261 **### REAL-FOOTAGE VALIDATION** + first ground-truth at-par ? over-merge fix confirmed; the
 gap is the rally-END **(2026-06-16)**
 262 The capstone of the day: the W1/W2/W1.5 windowing work, measured on **6 real**
 Kushagra/Yash clips (fully-local
 263 no-AI WASB, vs Gemini as a strong reference) and ? for the first time ? against **owner-**
hand-labeled ground truth
 264 on two of them. Full report:
`[`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md)` (adversarially
 265 verified ? a 17-agent synthesis killed 6/12 of its own draft claims). Reusable at-par
 analyzer: ``scratch/at_par.py``.
 266
 267 *** (1) The over-merge fix is real at scale*** (full-6, local-vs-Gemini, Gemini=reference
 not truth):
 268
 269 | aggregate (n=6 real clips) | baseline | W1 (cap=12) |
 270 |---|---|---|
 271 | windows (vs Gemini's 165) | 48 (mega-windows) | 250 |
 272 | mean window length | 62.9s | ***10.8s*** |
 273 | merge-groups (1 window ? 2 rallies) | 27 | ***10*** |
 274 | mean matched-IoU | 0.23 | 0.39 |
 275
 276 The signature pathology (GX010129: **2 windows / 267s / 27 rallies trapped**, mIoU 0.0)
 is gone. ***But W1 over-corrects***
 277 into over-segmentation (250 vs 165, ~1.5×) and it's **clip-content-driven**, NOT venue
 (mahadevpura-0 over-splits 1.55×
 278 like the GX clips ? the "GoPro-vs-court" story is unsupported at n=3/venue).
 279
 280 *** (2) First ground-truth at-par*** (vs human-labeled rallies; the honest scoreboard):
 281
 282 | clip (golden rallies) | metric | Gemini (cloud) | best LOCAL (no-AI) |
 283 |---|---|---|---|
 284 | mahadevpura-2 (40) | F1@0.5 / F1@0.25 | 0.557 / 0.835 | 0.296 / 0.722 (W1+W2+hardcap)
 |
 285 | GX010128 (52) | F1@0.5 / F1@0.25 | 0.813 / 0.857 | 0.457 / 0.756 (W1+W2) |
 286
 287 Local is ***~86%88%* of Gemini on rally DETECTION (F1@0.25) **** but only ***~53%56%* on tight
 boundaries (F1@0.5) ****. On
 288 GX010128 local even ***out-recalls Gemini*** ? it overlaps all 52 golden rallies; Gemini
 matched only 39/52 (missed 13).

```

289
290     **(3) ? The gap is the rally-END, confirmed at n=2 ? START is already at Gemini
parity.** Boundary error (matched
291     pairs, median seconds):
292
293     | clip | local \|?start\| | Gemini \|?start\| | local \|?end\| | Gemini \|?end\| |
294     |---|---|---|---|---|
295     | mahadevpura-2 | 1.74 | 1.46 | **3.12** | 1.19 |
296     | GX010128 | 0.78 | 0.98 | **2.12** | 0.76 |
297
298     Local nails the START (? or better than Gemini); the **entire** residual gap is the
rally-END over-extending (the
299     velocity-windowing ends at the last *in-flight* frame, not the last *sustained-active*
one). ? **next lever = a
300     sustained-inactivity rally-END rule (See et al.), NOT a generic boundary head and NOT a
serve/START classifier
301     (START is solved).**
302
303     **Lessons (for posterity):**
304     - **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet
F1@0.5 only 0.150 (W1+W2) ?
305     right count, wrong boundaries. Ground truth is the only honest judge; Gemini-count
agreement misled us twice.
306     - **Best windowing config is clip-dependent.** Hard-cap is best on continuously-active
clips (mahadevpura-1/-2) but
307     over-splits gap-rich clips (GX010128: W1+W2 0.457 > +hardcap 0.421). No single global
config; hard-cap is content-adaptive.
308     - **W2 (net-crossings) buys count by spending recall** ? on the real clips it dropped
~28 windows (~all on mahadevpura-0,
309     62743) for **6 net-new misses** and no ground-truth-backed gain (golden +0.019 n.s.).
**Stays default-OFF/retained, not promoted.**
310     - **tIoU@0.5 over-penalizes this task.** For a ~6s rally it tolerates only ~±1.9s and
the ~1?1.5s service window nearly
311     exhausts that ? it hides local's recall + START accuracy, and even flips config
rankings. The eval metric is now a
312     bottleneck ? **R2** (next): a **±2s tolerance-match F1 (?_start?2s, ?_end?1?1.5s) +
Hungarian 1:1 + an over-segmentation
313     guard + start/end boundary-error split**; tIoU/mIoU demoted to secondary.
314
315     **Regression discipline adopted (owner directive 2026-06-16):** maintain a **RAW eval
set** (currently-bad videos,
316     random-sampled) + a **GROWING GOLDEN-finalized eval set** (a video is promoted
raw?golden when an iteration brings it
317     *to par*) + a separate golden TRAINING set. **Every quality launch must not regress on
EITHER eval set** ? so a late win
318     can't silently undo an earlier one. The already-built ablation layer
(`backend/eval/ablation.py`) is the A/B mechanism;
319     wire the per-set aggregate into `run_regression.py` when R2 lands. (Privacy-safe, video-
free shipping of these sets =
320     the golden-regression-fixtures track.)
321
322     **Corpus: golden grew 6 ? 8** this session (`mahadevpura_2` 40 rallies, `GX010128` 52
rallies ? both real multicourt,
323     frame-aligned, added to `human_lovo_manifest.json`). **n=8 milestone numbers (LOVO,
tIoU-F1)** baseline 0.232 ? W1
324     0.403 ? W1+W2 0.419 ? **W1+W2+hardcap 0.437** (merge_rate 0.634 ? 0.000); milestone ?
baseline?W1+W2 = **+0.187 F1@0.5**.
325     All flags **default-OFF**; promotion to a shipped milestone is owner-gated and re-run at
the ~16-clip corpus.
326
327     ### Tier 1 (W1.5) ? hard-cap fallback: enforce the cap on continuously-active
trajectories *(2026-06-16)*
328     The real-footage 6-video local-vs-Gemini validation surfaced W1's one structural blind
spot: its
329     largest-internal-gap splitter **can't cap a window when the shuttle never stops moving**

```

```

(no inactivity
330     gap to cut at). On the real `mahadevpura-1` proxy, W1 left the 174 s blob as **2 windows
of ~87?102 s** ?
331     unwatchable, and 0% segmented vs Gemini's 9 rallies. **Fix (`window_hard_cap`, default-
OFF):** when an
332     over-long window has no qualifying internal gap, recursively chop it at its temporal
**midpoint** until
333     every piece ? `max_window_duration`. Still only cuts (never merges ? recall can't drop).
Threaded through
334     `trajectory_to_action_windows` ? `_split_overlong_groups(hard_cap=)`,
`WindowingPreset.hard_cap`,
335     `IndexingConfig.window_hard_cap`, served `trajectory_hybrid`, and `rally_seg_eval
--hard-cap/--vs-hard-cap`.
336
337     **MEASURED ? golden n=6 (W1 cap=12 ? +hard-cap):** ?F1 **+0.006** [?0.019, +0.033], **CI
spans 0,
338     sign-test 2/2, p=1.000 ? NOT significant** (+ seg_ratio +0.123: it adds some over-
split). **Honest read:
339     neutral on the clean golden set** ? the golden videos have few continuously-active >cap
windows, so the
340     fallback rarely fires. [[decision-rigor-norm]]: **not promotable on the golden number**
(CI spans 0).
341     **Real-footage value (the reason it exists), `mahadevpura-1` cached trajectory:**
342
343     | windowing | n windows | max window | mean window |
344     |---|---|---|---|
345     | baseline | 1 | 174.3s | 174.3s |
346     | W1 (cap=12) | 2 | 101.9s | 86.9s |
347     | **W1 + hard-cap** | **24** | **9.0s** | **7.2s** (? Gemini 5.7s) |
348
349     So the cap becomes a TRUE upper bound and the worst-case clip is finally segmented (24
windows vs Gemini's
350     9 ? it **over-segments**, since a blind midpoint chop is not rally-boundary-aware;
that's the honest
351     tradeoff and the motivation for the **boundary head / rally-state-aware split** (M1/R1),
the next lever).
352     **Status: retained default-OFF**, justified by the real-footage fix + the foundational
argument that
353     `max_window_duration` should be a true cap ? NOT by a golden lift (there is none at
n=6). Promote/flip with
354     more golden footage that actually contains continuous-motion blobs. Reproduce:
355     `rally_seg_eval --preset served --max-window-duration 12 --vs served --vs-hard-cap`.
Ties [[weak-signals-retained]].
356
357     ### Tier 1 (W2) ? "never-empty" safeguard makes the net-crossings gate production-safe
*(2026-06-16)*
358     Hardening that closes the one real-footage failure mode of W2 before the milestone flip.
**Found on real
359     footage:** on a noisy 4K-proxy trajectory (`mahadevpura-1`), W1 produced 2 windows but
the net-crossings
360     gate (mc=1) kept **0** ? the auto-estimated net axis + jittery shuttle track meant
neither piece registered
361     a clean crossing ? so the whole 174 s clip yielded **0 rallies (no reel)**. A gate that
can *zero out a
362     non-empty video* is not safe to layer onto the substrate. **Fix:** both gate sites
363     (`FusionSegmenter._select_for_handoff`, served; `rally_seg_eval.windows_for`, eval) now
fall back to the
364     **ungated** set whenever gating would drop *every* window of a video that had ?1. The
gate is therefore
365     strictly **pruning** ? it can shrink a candidate set but never zero it ? so W2-on-W1 is
**do-no-harm**: it
366     can only *match or beat* W1 per video, never strand a clip with no highlights.
**Verified:** golden delta
367     **unchanged** (W1 0.439 ? W1+W2 0.457, **+0.019**, 6/0, p=0.031 ? no golden video is
emptied at mc=1, so the

```

368 guard is a no-op there); on the real `mahadevpura-1` trajectory the guard turns the
 gate's `0` back into
 369 W1's 2 windows (`raw gate mc=1 = 0 kept ? guarded = 2`). 2 new tests
 370 (`test\_gate\_a\_never\_empties\_a\_video`, `windows\_for` mc=999 ? ungated). This makes
 **W1+W2 the safe milestone
 371 candidate** to flip on together.
 372
 373 ### Tier 1 (W2) ? net-crossings rally-state gate on W1: ?F1 +0.019, 6/0 *(2026-06-16)*
 374 The second Tier-1 increment, on top of W1. **Hypothesis (worth re-testing):** net-
 crossings (Gate-A) was a *null/negative*
 375 on the OLD coarse served windowing (it had nothing to clean ? the #151 revert), but W1's
 bounded windowing produces finer
 376 windows with over-split fragments (seg_ratio 1.44) ? exactly what a "did the shuttle
 actually cross the net?" gate should
 377 remove. **Confirmed.** **Shipped:** extended `rally\_seg\_eval` with an optional
 `min\_crossings` net-crossings gate
 378 (`rally\_gate.gate\_windows`) + a `--max-window-duration` CLI override, so the harness
 measures "windowing + rally-state
 379 gate". The gate itself is **existing code** (`FusionSegmenter.min\_crossings`); served
 path gets W1+W2 via config
 380 (`indexing.max\_window\_duration: 12` on `fusion\_hybrid` + `indexing.fusion.min\_crossings`:
 1) ? no new detector code.
 381
 382 **MEASURED on golden (n=6, LOV0; W1 cap=12, then + net-crossings ?1):**
 383
 384 | metric | W1 | W1 + Gate-A (mc=1) | ? |
 385 |---|---|---|---|
 386 | tIoU-F1@0.5 | 0.439 | **0.457** | **+0.019** (95% CI [+0.011,+0.027], **sign-test
 6/0**, p=0.031) |
 387 | F1@0.25 | 0.772 | 0.805 | +0.033 (6/0) |
 388 | segment_count_ratio | 1.44 | 1.33 | over-split trimmed |
 389
 390 Small but **statistically robust** (every video improves) and **served-measured** ? and
 a satisfying reversal of the
 391 Gate-A null: the cue only pays off *once the windowing is bounded* (signal value is
 windowing-dependent ? a generalisation
 392 of the eval?serve lesson). `mc=2` ? neutral, `mc=3` hurts (trims real rallies) ?
 recommend **mc=1**. Reproduce:
 393 `python -m backend.eval.rally_seg_eval --preset served --max-window-duration 12 --vs
 served --vs-min-crossings 1`.
 394 **Cumulative project number (golden n=6 aggregate tIoU-F1@0.5): 0.300 ? 0.439 (W1) ?
 0.457 (W2) = +0.157 total.**
 395 Default-OFF; promote via config. The richer multi-feature scorer (person/audio +
 direction-reversal/peak) is the W2
 396 follow-on (needs GPU person/audio feature re-extraction on the golden videos ? now
 feasible with the GPU grant).
 397
 398 ### Tier 1 (W1) ? bounded windowing fixes the over-merge: ?F1 +0.139, 6/0 *(2026-06-16)*
 399 The first quality *improvement* of the rally-quality project
 (RALLY_QUALITY_RESEARCH.md §6/§7,
 400 building on the Tier-0 baseline below). **Hypothesis:** the local segmenter doesn't miss
 rallies, it *fails to segment* ?
 401 "shuttle is moving" is treated as "rally in progress," so dead time + serves bridge
 rallies into mega-windows (Tier-0
 402 merge_rate 0.523; kushagra_singles = 2 windows for 32 rallies). **Shipped (default-OFF,
 config-gated):**
 403 `trajectory\_to\_action\_windows` gained `max\_window\_duration` + `min\_split\_gap` ? any
 window longer than the cap is
 404 recursively split at its largest internal inactivity gap (? min_split_gap; the most
 likely rally boundary, the See-et-al.
 405 sustained-inactivity spirit). It only **cuts**, never merges/extends, so recall cannot
 drop. Wired through `WindowingPreset`/
 406 `build\_windows` (eval) and `IndexingConfig.max\_window\_duration` ?
 `trajectory\_hybrid.process\_video` (served), default 0=OFF.
 407

```

408     **MEASURED on golden (n=6, LOV0, `rally_seg_eval --vs`, recommended cap=12s /
min_split_gap=0.5):**
409
410     | metric | baseline (Tier 0) | W1 (cap=12) | ? |
411     |---|---|---|
412     | tIoU-F1@0.5 | 0.300 | **0.439** | **0.139** (95% CI clears 0; **sign-test 6/0**,
p=0.031) |
413     | F1@0.25 | 0.501 | **0.772** | +0.271 |
414     | merge_rate | 0.523 | **0.038** | ?0.485 (over-merge essentially eliminated) |
415     | segment_count_ratio | 0.700 | 1.44 | now mildly over-*splits* ? W2's scorer to clean
up |
416
417     **Robust, not overfit:** every cap in 6?20 s clears 0 with 6/0 sign-test (the win isn't
a knife-edge); smaller caps trade
418     more over-split for more F1 (cap=6 ? +0.194 but seg_ratio 1.8). Picked cap=12 (rec-rally
upper bound) as the recommended
419     default; full sweep in the PR. **Eval?serve: this IS the served windowing + the cap, so
the win is served-measured ? no
420     proposal?served transfer risk** (unlike Gate-A). **Cumulative project number (golden n=6
aggregate tIoU-F1@0.5):
421     0.300 ? 0.439 (+0.139).** (An exact "% of the gap to Gemini closed" needs Gemini
measured on the *same* golden set +
422     metric ? a TODO; the existing 0.93-clean / 0.66-multi-court Gemini refs are per-venue,
not this aggregate, so don't mix
423     them.) Kept **default-OFF** pending owner go to flip the served default (one config
line); the residual over-split is the
424     explicit hand-off to **W2** (rally-state features into the scorer).
425
426     ### Tier 0 (E1) ? over-segmentation-aware eval harness + frozen local baseline
*(2026-06-16)*
427     First landed work of the **rally-quality project**
([RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) §7 ?
428     "bridge the local?Gemini gap"). **Why:** plain temporal-IoU F1 is nearly blind to the
*over-merge* the local
429     segmenter exhibits (one window swallowing many rallies), so we couldn't *measure* the
failure we set out to fix.
430     **Shipped:** `segmentation_metrics.merge_split_report` ? a bipartite overlap-graph
breakdown into
431     matched / **merge** (1 pred ? ?2 rallies) / split / missed / spurious +
`merge_rate`/`split_rate`, the
432     over-segmentation metric for this *binary* setting (the classic single-class edit score
is degenerate here ?
433     rally/background strictly alternate, so it collapses to the segment-count difference ?
and stays omitted);
434     plus `backend/eval/rally_seg_eval.py`, the one-command eval: golden manifest +
golden_features `gts` ? window
435     each cached trajectory under a `WindowingPreset` ? score (tIoU-F1 + F1@k + segment-count
+ merge/split,
436     bootstrap CI), and `--vs` diffs two windowings with a paired sign-test (the per-tier ?).
Pure/offline (CPU);
437     18 new tests; suite **722 green**, ruff + mypy clean.
438
439     **FROZEN BASELINE ? current local heuristic** (SERVED windowing 0.02/2.0/2.0, windows-
as-rallies, no AI; n=6 golden, IoU 0.5):
440
441     | metric | mean [95% CI] |
442     |---|---|
443     | tIoU-F1@0.5 | **0.300** [0.119, 0.494] |
444     | F1@0.25 | 0.501 [0.239, 0.747] |
445     | **merge_rate** | **0.523** [0.258, 0.788] |
446     | segment_count_ratio | 0.700 [0.405, 0.957] |
447
448     The `0.300` **matches the independent served-side measurement below** (min_crossings=0 =
`wasb_hybrid` = 0.300) ? cross-validated.
449     **Tier-0 "number" = this baseline** (Tier 0 establishes the ruler; it does not change

```

quality). The over-merge is now a metric:

```

450     **merge_rate 0.523** ? 52% of golden rallies swallowed into a merge group
(kushagra_singles = 1.00, just **2 windows for 32 rallies**;
```

451 gbaaddy 0.85; mahadevpura 0.52). Every later tier's delta is measured against this via
`python -m backend.eval.rally\_seg\_eval --vs ?`;

452 per-tier + cumulative deltas land here.

453

454 **### Served-side Gate-A measurement on the LIVE FusionSegmenter ? the independent run**
behind the #151 revert *(2026-06-14)*

455 The owner-gated follow-up flagged by the (now ****REVERTED****) SERVED-DEFAULT entry below.
An ****independent**** served-side check ?

456 separate from, and corroborating, the [#148](https://github.com/avidullu/badminton-
highlight-indexer/pull/148) litmus + #151's

457 run ? of whether the Gate-A default flip (PR #146) regresses the live path. ****Verdict:**
it does ? the flip was reverted in

458 [#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151); this entry is
the detailed measurement. ****Driven the**

459 ****real**** served seams (`\_enrich\_candidate\_stats` ? `net\_crossings`,
`\_select\_for\_handoff` ? the gate) over each golden video's

460 ****cached**** WASB trajectory, at the ****production windowing**** `process\_video` actually
uses

461 (`velocity\_thresh=0.02, merge\_gap=2.0, min\_window=2.0`) ? no WASB re-run (the runner has
no cache-hit early-return), no GPU, no

462 Gemini. Config loaded exactly as production (`load\_config()`); segmenters from the
registry. Repro:

463 `scratch/served\_gateA\_verify.py` + `scratch/served\_gateA\_sweep.py` (artifacts in
`output/gateA\_served\_verify/`); the standing

464 productionized guard is `backend/eval/served\_gate\_a.py` (#148).

465

466 - **** (a) No-regression invariant CONFIRMED on real served code. **** At `min\_crossings = 0`,
`fusion\_hybrid`'s seeding ****and****

467 handoff set are ****byte-identical to `wasb\_hybrid`**** on all 6 golden videos (same
candidate windows; handoff = all windows =

468 the base seam). The "fusion adds nothing when the gates are off" property is now
verified on the **live** served path, not

469 only by code review. ?

470 - **** (b) The +0.074 win does NOT transfer to the served windowing. **** The offline
promotion (+0.074, 6/6, p=0.031) was measured

471 on the eval harness's ****recall-oriented *proposal* windowing**** `(0.005, 1.0, 0.5)`,
where CONTROL ****over-segments****

472 (segment-count ratio 1.68) and Gate-A's job is to clean up that fragmentation (?
1.01). The ****served path uses a coarse**

473 ****windowing**** `(0.02, 2.0, 2.0)` that already merges aggressively (low over-
segmentation), so Gate-A has little FP to remove

474 and instead trims ****real**** rallies whose WASB trajectory registered only 0?1 net
crossings. ****Candidate-handoff-level F1**

475 (pre-Gemini, IoU ? 0.5) on the served windowing, n=6:******

476

	min_crossings	mean F1	precision	recall	real-rally windows lost vs off	?F1 vs off
0 (Gate-A OFF = `wasb_hybrid`)	0.300	0.317	0.290	0	?	
1 **0.303**	0.327	0.288	1	**+0.003**		
2 (shipped default, PR #146)	0.292	0.337	0.261	**12**	**?0.008**	
3	0.269	0.338	0.226	28	?0.031	

483

484 At the shipped `mc=2`, mean F1 goes ****0.300 ? 0.292 (?0.008)**** and Gate-A drops ****12**
windows that match golden rallies

485 (net_crossings almost all exactly 1.0), concentrated in doubles (`targetest\_doubles`
?0.051). All deltas are ****tiny / within**

486 n=6 noise, so this is not a catastrophic regression ? but `mc=2` is ****not**** a
served-side **improvement**, and the

487 held-shuttle "guard" is removing genuine rallies, not just feeds.

488 - ****Served relevance (honest). **** A dropped candidate window ****never reaches Gemini****, so
each lost real rally is an

489 **unrecoverable served recall loss**; the precision "gain", by contrast, is partly
 what Gemini already does at the handoff ?
 490 so the served-side tradeoff is at best neutral and plausibly net-negative, i.e. the
 candidate-level ≈ 0.008 likely
 491 *understates* the served downside.
 492 - **Root cause = eval ? serve windowing.** The harness promotes/refuses variants on a
 windowing the production path does not
 493 serve. Gate-A's value is conditional on over-segmentation; the coarse served windowing
 doesn't over-segment, so the gate
 494 inverts. [[decision-rigor-norm]]: the foundational issue (harness windowing ? served
 windowing) outweighs the single
 495 proposal-windowing number.
 496
 497 **Outcome ? REVERTED (see the entry below).**
 [#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)
 498 restored the pre-#146 state: `config.json` `indexing.default\_segementer` ? `gemini` (the
 `indexing.fusion` block removed) and
 499 `FusionConfig.min\_crossings` default **2 ? 0** ? Gate A is now **OFF by default but
 still opt-in**
 500 (`indexing.fusion.min\_crossings: 2`), the persisted-superset invariant intact. Our
 independent numbers match #151's (mean
 501 served $F1 \approx 0.008$, ~12 rallies dropped, 2/6 losses). If Gate A is ever opted back in on
 the served path, `mc=1` is the
 502 least-bad operating point (?flat $F1$, 1 rally lost). **Open follow-up:** **align the
 experiment-harness windowing with the
 503 served windowing** (or re-run promotions on the production windowing) so a harness lift
 actually predicts served behaviour ?
 504 the real lesson here. The offline ablation + unit tests remain valid for what they
 measured.
 505
 506 ### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
 feature *(2026-06-14)*
 507 The owner adopted the **HYBRID** posture: personalization as a **flag-gated FEATURE on a
 generalization-first baseline**,
 508 paired with a **contribution flywheel** so per-user gains and the shared baseline
 compound together (full design in
 509 PERSONALIZATION_PLAN.md, merged
 [#144](https://github.com/avidullu/badminton-highlight-indexer/pull/144)).
 510 This checkpoint landed the three **honest-eval RAILS** that keep a per-user/small-N
 tuning layer from promoting noise or
 511 silently regressing a user:
 512
 513 - **Per-user promotion gate ? [#141](https://github.com/avidullu/badminton-highlight-
 indexer/pull/141)**
 514 (`backend/eval/per\_user\_gate.py`): a **min-N floor** + **structural-guard exemption**
 ? per-user/small-N tuning can't
 515 promote noise and can't disable a structural guard (the proven Gate-A held-shuttle
 guard stays on regardless of a thin
 516 user corpus).
 517 - **Held-out-USER learning curve ? [#142](https://github.com/avidullu/badminton-
 highlight-indexer/pull/142)**
 518 (`backend/eval/per\_user\_eval.py`): turns the vague "<K videos to superior quality"
 claim into a **falsifiable release
 519 number** (the `smallest\_k\_with\_lift` measured on held-out users).
 520 - **Per-user no-regression guard ? [#143](https://github.com/avidullu/badminton-
 highlight-indexer/pull/143)**: a
 521 retrain/baseline-upgrade can **never make a user worse than their incumbent** (frozen
 per-user CONTROL; a regressing
 522 candidate is refused).
 523
 524 **Read HONESTLY:** these are the **eval/safety rails** for the personalization feature ?
 not new measured
 525 rally-detection wins. No $F1$ moved this checkpoint; what landed is the discipline that
 lets per-user tuning ship without
 526 regressing the baseline or over-claiming on $n \approx 6$ -per-user data.

```

527
528     **Locked owner decisions (2026-06-14):** identity = *design for the north-star (multi-
user/cloud), implement for current
529     use (single-user local)*; the inline per-user classifier sits **behind a cloud-load
interface** (local store = one impl);
530     **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate** (a one-
video user is ALWAYS fully segmented by
531     the baseline batteries); **free-tier videos ARE pooled** into the shared baseline
(consent baked into free terms,
532     minors-exclusion enforced at the pooling query); and the previously-pending structural-
guard sub-decision is now LOCKED ?
533     **Gate-A-only structural + `min_n = 5`**.
534
535     ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
536     The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
537     replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign
538     test; never a bare mean at n=6)**:
539
540     | Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
541     |---|---|---|---|---|---|
542     | shuttle-only | 0.307 | ? | ? | ? | ? |
543     | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |
**No** |
544     | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
545
546     **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
547     default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?
`players_in_court` counts spectators);
548     per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
549     (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**
shuttle-only (0.307) ? but n=6 can't
550     confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and
measure audio on shuttle-only
551     *directly* (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
552     results are kept on purpose ? this is the honest ablation backbone for the paper aim.**
553
554     ### Gate-A served-default flip ? **REVERTED (eval?serve regression)** *(2026-06-14)*
555     **The #146 flip of Gate-A (net-crossings ? 2) to the live served default was REVERTED.**
A served-side check ?
556     the repeatable litmus in [#148](https://github.com/avidullu/badminton-highlight-
indexer/pull/148)
557     (`backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py`),
corroborated by an independent
558     served run ? found Gate A is **not a win on the served path**: on the production
(COARSE) windowing it is
559     **neutral-to-negative, mean served ?F1 ?0.008**, dropping ~12 real rallies across the
golden 6 (2 losses, 0
560     clear wins). It is a **confirmed served regression**, so the owner directed a revert to
restore the known-good
561     state.
562
563     **Why eval ? serve (the honest root cause).** The +0.074 below is a property of the
**offline harness's
564     permissive PROPOSAL windowing** (`velocity_threshold=0.005, merge_gap=1.0,
min_window=0.5`), which emits many
565     short candidate windows ? leaving abundant held-shuttle junk for Gate A to cut. The
**served COARSE windowing**

```

```

566     (`velocity_threshold=0.02, merge_gap=2.0, min_window=2.0`) already merges flight bursts
into a few long windows
567     and drops short ones (served over-seg ratio already  $\ll 1$ , i.e. under-segmentation),
so there is little junk
568     left to cut and the gate instead occasionally trims a borderline true window ? slightly
negative ?F1. The gate
569     *itself* is correct (#148's control test reproduced  $+0.074 / 6\text{-of-}6 / p=0.031$  exactly
under proposal
570     windowing **on the original-6 corpus** ? archived; see the re-grounding note below); the
**operating point is
571     what differs**. This is the canonical eval?serve trap.
572
573     **What the revert restores** (back to the pre-#146 no-regression state):
574     - `config.json` ? `indexing.default_segementer` **`fusion_hybrid` ? `gemini`** (the prior
served detector); the
575     explicit `indexing.fusion` block added by #146 is removed.
576     - `backend/config/models.py` ? `FusionConfig.min_crossings` default **2 ? 0** (Gate A
**OFF** by default).
577     - Tests (`tests/test_config.py`, `tests/test_fusion_hybrid.py`) restored to the Gate-A-
OFF defaults; the
578     **persisted-superset invariant is preserved** (gating only ever filtered the AI
handoff, never persistence).
579     - `eval_baselines/experiment_leaderboard.json` ? the #146 `promotion` record carries a
`reverted (eval?serve)`
580     note.
581
582     **Gate A is NOT deleted ? it remains opt-in** (`indexing.fusion.min_crossings: 2`) for
footage with more
583     held-shuttle/service-feed false positives than the golden 6, and the offline ablation
evidence below stands
584     (it is a real proposal-windowing win). The path back to serving it is to **align the
served windowing toward
585     the proposal operating point** and re-run the #148 litmus to confirm the lift transfers.
The #148 litmus is
586     now the standing guard: it pins the honest measured behaviour on the **present golden
corpus** and trips if a
587     future windowing/corpus change silently shifts it.
588
589     > **Re-grounded 2026-06-18.** The original-6 single-court golden clips this litmus was
calibrated on have been
590     > retired from the box (their labels are gone); the present golden set is **9 multicourt
clips**. On that
591     > recall-starved corpus (WASB tracks one shuttle while several courts rally ? served
over-seg ratio  $< 1$ ) the
592     > served Gate-A ?F1 is now **0.034** (was 0.008), and under the permissive proposal
windowing the gate is
593     > **~F1-neutral** (?F1 ? 0.001) while still cutting over-seg (1.16?0.64) ? it no longer
reproduces the +0.074
594     > live. The canonical **+0.074 / 6-of-6 / p=0.031** win remains a real result of the
original-6 corpus,
595     > preserved in the [first-A/B archive](archives/quality-iterations/2026-06-first-ab-
experiment/README.md).
596
597     ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
598     **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The
measured win
599     (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the
600     [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest

```

11. assistant (2026-06-19T15:47:04.066Z)

I found the key stale line in QUALITY_ITERATIONS.md at line 18-21: "Golden corpus = the 6

labelled matches". Note the manifest names confirm `Badminton\_BXH\_2` (underscore) and `Boxhill\_Doubles`, `GX020094` are entries 11, 14, 15. Let me note that line 592-593 says "the present golden set is **9 multicourt clips**" in a re-grounding note dated 2026-06-18 ? I need to check if this is the present-corpus claim or a dated historical note. It's a dated `Re-grounded 2026-06-18` note but it makes a present-tense CURRENT claim ("the present golden set is 9 multicourt clips"). This contradicts n=15. Let me also note line 62 already says "n=15 golden". Let me read the rest of QUALITY_ITERATIONS.md and check the other docs.

12. user (2026-06-19T15:47:05.487Z)

```
600    [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest
601    bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
602    folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the
recall-oriented *proposal*
603    candidate set; flipping the **live served default** stayed owner-gated at the time of
this entry. **Update: the
604    flip was attempted (#146) and then REVERTED** ? on the served COARSE windowing Gate A is
neutral/negative
605    (?0.008), an eval?serve regression; see the REVERTED entry above. Gate A stays opt-in
606    (`indexing.fusion.min_crossings: 2`), default OFF. **Corroborated on real footage:**
offline
607    re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated
`output/GX010125_run/`) ? 45 candidates ?
608    CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7
min)** of likely
609    held-shuttle/feed FPs (no golden labels for this footage ? divergence, not F1).
610
611    **Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130).** The
612    first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added
613    additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train**
fold) + `calibrate`
614    (Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423
(+0.116)**; testlarge
615    0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)**; Threshold-in-LOVO
is the big lever;
616    calibration is available too.
617
618    **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
619    ([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are
self-contained.
620
621    ### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
622    First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB
trajectories, no GPU).
623    **Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test
**p = 0.031**, CI
624    [+0.042, +0.104], over-segmentation (segment-count ratio) **1.68 ? 1.01**. The window-
level 5-feature logistic
625    **did not clear the bar** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the
stronger per-frame
626    prototype). The framework held up: **C1 promoted one variant and refused another; C4
surfaced over-segmentation
627    tIoU hid.** Full record + tables ?
628    [archives/quality-iterations/2026-06-first-ab-experiment/](archives/quality-
iterations/2026-06-first-ab-experiment/README.md).
629    Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure
is the C2 (calibration)
630    + threshold-in-loop task list.
631
632    ### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF
*(2026-06-14)*
```

```

633     **Why.** A multi-agent literature review (folded into
634     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13) confirmed the
framework's *skeleton* is
635     standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion
? cite, don't claim),
636     and flagged that our **measurement** is walking into known small-N traps that would make
"promote only on
637     measured lift" promote noise or hide a real win. Four corrections, shipped as **pure,
additive helpers** (no
638     existing `compare`/LOVO`/metrics.score` behaviour changed ? callers opt in):
639     - **C1 ? never a single mean.** `backend/eval/eval_stats.py`:
`mean_with_ci`/`bootstrap_ci` (deterministic,
640     seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*"
test the corpus already
641     needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean
is not trustworthy.
642     - **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt +
isotonic (PAVA) calibrators
643     + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an
integer net-crossing
644     count are comparable before the arbiter and the over-confident cue can't swamp the
rest.
645     - **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-
one-group-out by
646     match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
647     - **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
648     (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
649     is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler
exists to fix.
650
651     **Status.** Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free
lanes); ruff + mypy clean.
652     **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to
report CIs + F1@k is
653     the follow-on, owner-gated). The discipline they encode is the methodology for every
future cue measurement ?
654     see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
655
656     ### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
657     **Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A**
(`min_crossings?2`)
658     and a person-occupancy **Gate B** raises **precision** on the held-shuttle / warm-up /
one-sided-feed /
659     spectator false positives ? *without* hurting recall, because the shuttle stays the
primary,
660     camera-invariant seed.
661
662     **Scope ? and an honesty boundary from the archive (read this).** The
663     [`2026-06-multivideo-lovo`](archives/quality-iterations/2026-06-multivideo-
lovo/README.md) iteration
664     already smoke-tested **court-region detection as a fix for the *multi-court contrast*
confound** and
665     found it **low-headroom**: only 12?14% of dead-time shuttle detections fall *outside*
the foreground
666     court, so a perfect court gate lifts contrast to only ~1.6?2.2× (below the 3.6× "works"
threshold) ? the
667     multi-court problem is **temporal**, and the **learned model**, not a court mask, is its
answer. So Gate B
668     here is **NOT** sold as the multi-court fix. It targets a **different, previously-
untested** failure mode
669     (held-shuttle/warm-up/one-sided/spectator FPs). Two distinct levers; don't conflate
them.

```

```

670
671     **Data support for Gate A.** The same archive shows real rallies on the multi-court
matches contain
672     **?2 net-crossings 97% of the time (median 5)** while a single service feed crosses
~once ? `min_crossings?2`
673     is a principled, data-backed discriminator (it's the held-shuttle owner-bug killer).
674
675     **Shipping shape.** A NEW `fusion` segmenter, **default OFF** (`default_segmenter`
unchanged); Gate A
676     wired into the live trajectory path behind `indexing.min_crossings` (default 0 = OFF ?
wasb/tracknet
677     byte-identical today); the court mask is an **optional** config polygon (omit ? Gate B
is
678     player-count-only; supply ? masked). Person features (`players_in_court,
two_sided_motion,
679     mean_player_motion, in_court_ratio, shuttle_player_colocation`) persisted into candidate
`stats` for P3's
680     learned fusion. Reuses `PersonDetector` (P1) + `rally_gate` (already existed) ? no
reinvention.
681
682     **Status (2026-06-14): SHIPPED, default-OFF.** `FusionSegmenter`
(`backend/pipeline/segmenters/fusion_hybrid.py`,
683     registry key `fusion_hybrid`) + the pure feature math
(`backend/pipeline/detectors/fusion_features.py`) + two
684     identity hooks on the trajectory engine (`_enrich_candidate_stats`,
`_select_for_handoff`) + `PersonDetector`.
685     detections_per_frame` + the `FusionConfig` block. Built design-first (a
`fusion-p2-design` fan-out workflow), then
686     **adversarially reviewed by a 31-agent workflow (25 findings ? 19 confirmed)**: the
headline confirmation is
687     **NO-REGRESSION verified** ? wasb/tracknet twins persist byte-identical stats and hand
off all windows (pinned by
688     the extended equivalence test); default `FusionSegmenter` == its wasb substrate. The
confirmed low/medium findings
689     were fixed (frame-dim guards, `round()` frame indices, the per-window axis-estimate
redundancy ? a reusable
690     `rally_gate` estimate, honest torch-import docstrings, and 6 new tests closing the
person-detector / Gate-B /
691     substrate / colocation-halo gaps). Suite 441?467 green**; ruff + mypy clean.
692     **Two findings deferred with intent** (reviewer-agreed): person features are computed
eagerly even for
693     Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics);
and `fusion.velocity_threshold`
694     must be set equal to the substrate's to A/B against a tuned wasb run (documented in
`FusionConfig`).
695     **Lift still to be measured** offline on the 6 golden videos via
`experiment.compare(CONTROL, fusion)`; the
696     held-shuttle-drop + person-feature values are GPU/owner-side. *(Measured numbers
appended here when run ? no claims before.)*
697
698     ### Person cue extracted to a reusable producer (fusion P1) ?
[#115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13
699     Lifted the torchvision detector + ByteTrack + velocity windowing out of `yolo_hybrid`
into
700     `backend/pipeline/detectors/person_detector.py::PersonDetector`. **No behavior change**
(proven by the
701     unchanged `process_video` output tests); `yolo_hybrid` now orchestrates only. **Why it's
a quality move:**
702     it makes "add the person cue to fusion" an *add-a-producer*, not a segmenter rewrite.
Suite 440?441 green.
703
704     ### Experiment harness: A/B without rollback risk ?
[#114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13
705     Shipped `backend/eval/experiment.py` ? the thin variant/experiment layer (the audit's
"-80% already

```

```

706     exists" held; pure glue over
`metrics`/`calibration`/`classifier`/`training`/`regression.gt_hash`). Gives
707     `Variant` (+ pinned `CONTROL`), `compare()` (labeled A/B, LOVO-honest),
**`compare_unlabeled()`** (the SxS
708     read on NEW unlabeled footage ? agreed / A-only / B-only), and a JSON experiment
leaderboard. **This is
709     the instrument every iteration is measured on.** 19 tests incl. the no-regression
invariant (all cues OFF
710     ? byte-identical to control). Suite 421?440 green.
711
712     ---
713
714     ## Archived predecessors (the historical trail)
715     Completed iterations live under [archives/quality-iterations/](archives/quality-
iterations/README.md):
716     - **`2026-06-initial`** ? seed of the quality phase.
717     - **`2026-06-human-gt-tuning`** ? first eval vs **human** GT on a partial-label video;
partial-label
718     harness + labeled-window methodology; baseline 0.650 ? tuned 0.742 (*fit-on-self*);
found+fixed a
719     negative-start ingestion bug (collector PR #13).
720     - **`2026-06-multivideo-lovo`** ? the **n=6 corpus + honest LOVO** ; established the
heuristic is
721     **footage-fragile** (multi-court Kushagra 0.16 / gBaaddy 0.28), the ***"contrast"
metric** predicts
722     heuristic F1 monotonically, and a **learned per-frame prototype overtook the heuristic
at n=3 and holds
723     ~+0.10** (n=6: learned 0.583 vs heuristic 0.480), with **AUC 0.83?0.92 decoupled from
contrast**. This
724     is the data-backed green light for the owned model.
725
726     ---
727
728     ## Standing lessons (carry-forward)
729     - **Persist detection once, re-score variants forever.** Expensive/risky = per-frame
detection (GPU/WSL);
730     cheap/swappable = the decision. Most experiments never enter the served pipeline.
731     - **LOVO or it didn't happen.** Score on a held-out *video*, never a held-out window
from the same video.
732     Group by **match**, not file. Sign test needs n ? ~n=10/9-wins for p<0.05.
733     - **The multi-court confound is TEMPORAL, not spatial/loudness** ? three heuristics were
tested and failed
734     for the same reason (foreground spatial gate **?0.04** ; audio energy **AUC ~0.6** ;
court-region
735     separability **12?14% out-of-region**): the dead-time confusion is rally-vs-non-rally
activity in the
736     *same place and frequency band*. Only the learned model's temporal/structural features
crack it (AUC
737     0.83 on Kushagra/gBaaddy). **Don't re-derive these dead ends.**
738     - **Court-masking ? multi-court fix** (see above) ? it's a **precision** tool for off-
court persons and
739     one-sided feeds, a *different* lever; measure its lift, don't assume it.
740     - **Net-crossings are a strong rally signal** ? 97% of real rallies cross ?2x (median
5); single feeds ~1x.
741     Gate A leans on this.
742     - **Small-N feature discipline.** Few, scale/translation-invariant features
(counts/ratios, not raw
743     coords) + a logistic regression, not a net ? raw coords memorize this court/camera.
744     - **Calibration, not representation, is the small-N wall** ? learned probabilities
separate (high AUC) but
745     the operating point doesn't transfer with few training videos. More distinct matches
is the highest-ROI
746     unblock.
747     - **Gate A already existed, just wasn't wired to production** ? the cheapest precision
win is plumbing, not

```

748 new capability. Always check what the eval drivers already do that the live path
doesn't.
749

13. user (2026-06-19T15:47:05.691Z)

```
1      # Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)
2
3      > **Status:** `DRAFT` ? implementation **owner-gated** . **Owner:** Avi . **Created:**
2026-06-18 . **Last updated:** 2026-06-18
4      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; pointer in memory
`current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
6      > **Relation to existing docs:** **peer-of**
[ `RALLY_QUALITY_RESEARCH.md` ] (RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework
doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on**
[ `MULTI_SIGNAL_FUSION_PLAN.md` ] (MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics),
[ `QUALITY_ITERATIONS.md` ] (QUALITY_ITERATIONS.md) (the R-log / empirical backbone),
[ `GOLDEN_VIDEOS.md` ] (GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.
7      > **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) /
`[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read
$6.
8
9      ---
10
11     ## 0. TL;DR
12     We are using the **growing golden corpus** (human rally labels) to find and close the
rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip
already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a
real rally, its edges are tight ( $|?| < 1$  s). **The remaining gap is DETECTION, not boundaries:**
(G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision**
? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-
extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity
alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the
priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall**
(opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1
lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the
player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style
precision) as the durable win. **Caveat:** two golden clips so far (n=13 corpus,
GX010141+GX010137) ? directional, not conclusive.
13
14     ## 1. Problem & goal
15     The detector is good enough to ship a highlights product, but the golden eval shows
clip-dependent accuracy that we want to **close consistently on every video, not just on
average** ? catching the failure modes in the quality loop instead of after launch.
16
17     - **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs
the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set
discipline, [[eval-dual-set-regression]]).
18     - **"Good" =** the three gaps below closed enough that a milestone flip (default-ON)
clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report
([[launch-report-convention]]).
19     - **Read first to get current:**
[ `RALLY_QUALITY_RESEARCH.md` ] (RALLY_QUALITY_RESEARCH.md) $1?2 (the gap + diagnosis),
[ `MULTI_SIGNAL_FUSION_PLAN.md` ] (MULTI_SIGNAL_FUSION_PLAN.md) $4 (in-flight vs in-hand),
[ `HOW_RALLY_DETECTION_WORKS.md` ] (HOW_RALLY_DETECTION_WORKS.md) (the 2-layer model).
20
21     ## 2. Decisions locked
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | The gap is **DETECTION (recall + precision + rally-END), not boundary precision**
| golden GX010141, 2026-06-18 `[verified]` | Don't build a generic boundary head; spend on
rally-state signals. |
25     | D2 | Measure **per-video AND aggregate; no regression on ANY video** | [[eval-dual-
```



```

set-regression]] / [[decision-rigor-norm]] | A win must broadly help, not trade one clip for
another. |
26 | D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus |
Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |
27 | D4 | New signals land **default-OFF + flag-gated**, promoted only on measured lift |
[EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) / [[weak-signals-retained]] | Zero-regression
rollout; retained even on a small-n null. |
28 | D5 | **Rally-END needs a PLAYER-state signal** ? shuttle velocity can't separate a
competitive rally from a casual post-point return | owner insight + R4 limitation, 2026-06-18
[design] | Add person kinematics / shuttle-in-hand as the rally-end discriminator. |
29
30 ## 3. Foundation ? the measured gaps (golden ground truth)
31 Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone
windowing = cap6 + R4 ([QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) R3/R4). Local & Gemini
vs truth, n=2 so far: [verified]
32
33 | clip (golden) | system | found | tIoU@0.5 P/R/F1 | tolF1 | hits | false+ | missed |
\|?start\| | \|?end\| |
34 |---|---|---|---|---|---|---|---|---|
35 | **GX010141** (29, hard/drilling) | LOCAL | 22 | 0.59/0.45/**0.51** | 0.43 | 13 | 11 |
16 | 0.78 | 0.68 |
36 | | GEMINI | 16 | 0.94/0.52/**0.67** | 0.53 | 15 | 4 | 14 | 0.31 | 0.46 |
37 | **GX010137** (34, normal match) | LOCAL | 39 | 0.77/**0.88**/**0.82** | 0.68 | 30 | 14
| 9 | 0.39 | 0.44 |
38 | | GEMINI | 27 | 1.00/0.79/**0.89** | 0.89 | 27 | 0 | 7 | 0.50 | 0.52 |
39
40 - **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25
0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above
the mean. [verified]
41 - **Opposite failure modes ? the central finding (n=2).** **LOCAL = high recall, low
precision; GEMINI = high precision, low recall.** On the normal clip GX010137, local's **recall
0.88 BEATS Gemini's 0.79** (local finds *more* real rallies) but local fires 14 false positives;
Gemini fires **zero** false positives but misses 7 rallies. They are **complementary** (on
GX010141 they overlap on only ~3 rallies yet each hits ~13?15 of the golden set). ? **fusion
(local's recall + Gemini-style precision filtering) is the clear win.**
42 - **The gaps, re-prioritized by the data:**
43 - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39
false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter
local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-
recalls it.
44 - **G1 ? recall** is clip-dependent: a local *strength* on normal play (? Gemini), but
on hard continuous-drilling clips *both* miss ~half (GX010141 recall ~45?52%). The recall lever
matters mainly for the hard clips.
45 - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ?
window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false
positives/merges.
46 - **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ?
normal clip ??0.07 (0.82 vs 0.89), and local *out-recalls* Gemini there. The deficit is
precision, not detection.
47 - **What is NOT the gap:** boundary precision. When local hits a rally its edges are
tight (|?start|/?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing
is doing its job. [verified]
48 - **Long-rally lens refinement (P7, n=15, [verified]):** filtering the eval to >5s
rallies (the eye-catching highlights) shows local's single-court over-fire is **short-FP-
dominated** (seg_ratio 1.43?1.04 ? duration-filterable), but **multicourt still over-fires 1.58x
on LONG rallies** (adjacent-court games are genuine long rallies). ? the multicourt precision
gap needs **court-localization (Gate-B / court mask, Lever C)**, NOT a duration filter; and the
strict tolerance precision is gated by **over-merge contamination** of the long-pred set, not
just short FPs. See [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) "P7".
49 - ? **Data caveat:** GX010141 was labeled before ~3 rally-annotator-tool bugs were found
+ fixed (GX010137 is post-fix). GX010141's numbers may shift on a re-export ? re-verify before
drawing hard conclusions from it.
50
51 ## 4. Design / architecture ? the levers (REUSE MAP)
52 The signal mechanics live in

```

```
[`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md); this is which lever attacks which
gap.
53
54 - **Lever A ? player-state rally-END rule** (targets **G3 + G2**) `[design]`. Generalize
R4 from shuttle-only to **shuttle OR bilateral player motion**: end the rally at the last frame
with competitive evidence from *either*, and **hard-terminate on a "shuttle picked up" event**.
Two cheap detectors, no new model required:
55 - *Player kinematics* ? person velocity drops / players go stationary after a point
(the casual returner taps it while the other walks). **Reuse:** the person cue producer
(torchvision + ByteTrack) in `yolo_hybrid` (fusion P1, `[verified]` exists).
56 - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation` (shuttle box ? person
box = held) `[design]` from the fusion plan $4, **or** the no-new-model proxy: **sustained WASB
`Visibility=0`** after the last fast stroke (held shuttle goes dark). Use it as a **termination
anchor + reverse-timeline search** to the true end (last net-crossing / last fast-shuttle / last
bilateral motion) ? the owner's idea, refined: the hold event is *late*, so anchor-then-
backtrack, don't use its timestamp as the end.
57 - **Lever B ? recall** (targets **G1**) `[design]`. Both systems miss real rallies ?
detector sensitivity: proposal-windowing / velocity-threshold tuning, and the **complementary-
fusion** angle below. Measure recall floor per video; the highest-leverage labels are the
*missed* rallies (active learning, RALLY_QUALITY_RESEARCH $4.3).
58 - **Lever C ? signal/ensemble fusion** (targets **G1**) `[design]`. Local & Gemini (and
shuttle vs person cues) hit *different* rallies ? a union/ensemble lifts recall. Already-built
scaffolding: `rally_gate.py` **Gate A** (net-crossings, structural rally-state, `[verified]`
built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature
path.
59 - **Pure-precision cleanup** (**G2**): `rally_gate` Gate A kills single-feed / held-
shuttle *starts* structurally (?0?1 net crossings vs a real rally's several) ? the cheapest G2
win and independent of the person stack.
60
61 **Reuse map (what's new vs there):** `rally_gate.py::gate_windows` (Gate A, built) .
`tracknet_runner` R4 end-trim (extend, don't replace) . `backend/eval/windowing.py` (preset
knobs) . person producer in `yolo_hybrid` (built) . `shuttle_player_colocation` (new feature) .
`scratch/ingest_golden.py` + `scratch/boxhill17_score_golden.py` (the per-video golden scoring
loop, `[verified]`).
62
63 ## 6. Honest limits ? what this does NOT do
64 - **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with
several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 /
GX020140 / a large-count clip next).
65 - **Nothing here is shipped or even started** ? implementation is owner-gated. A DONE
row in $7 means *measured on golden*, not *flipped on by default*.
66 - **Player-state levers need GPU person-detection on the golden clips** (not yet
extracted); until then Lever A/B are design, not data.
67 - The boundary tightness is from clips where local *hits* the rally; it says nothing
about the rallies it misses (G1).
68
69 ## 7. Deliverables & progress tracker ? **source of truth**
70 Legend: ? Todo . ? In progress . ? Done . ? Gated. One small PR per row, off
`origin/master`, no stacking.
71
72 | ID | Deliverable | Depends on | Gated? | Status | PR |
73 |----|-----|-----|-----|-----|----|
74 | P0 | Golden scoring loop (`ingest_golden.py` + `boxhill17_score_golden.py`); GX010141
ingested (#12) + scored | ? | No | ? | this work |
75 | P1 | Ingest the rest of the Boxhill golden batch (137/020140 + a large-count clip) ?
per-video baseline table | owner labels | ? owner | ? | ? |
76 | P2 | Wire `rally_gate` **Gate A** (net-crossings) to the served path ? cheapest G2 win
| ? | ? owner | ? | ? |
77 | P3 | Extract person-velocity + `shuttle_player_colocation` on the golden clips
(default-OFF features) | P1 | ? owner | ? | ? |
78 | P4 | **Lever A**: player-state rally-END rule (R-next) ? measure ?F1 / ?merge-rate /
?\\?end\\ vs R4 | P3 | ? owner | ? | ? |
79 | P5 | **Lever B/C**: recall + complementary fusion ? measure recall lift, no precision
regression | P3 | ? owner | ? | ? |
80 | P6 | Launch decision: milestone + best levers default-ON, with a Launch Report (both
```

```

rulers + over-seg guard) | P4,P5 | ? owner | ? | ? |
81 | P7 | **Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel
knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-
fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58x even on long
rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-
contaminated). See [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) "P7". | ? | No (eval+cfg,
default-OFF) | ? measured | this PR |
82
83 ## 8. Open questions ? owner / external
84 1. **Which lever first** once labels land ? the no-model wins (Gate A wiring + shuttle-
invisibility end-anchor), or go straight to person-velocity? *(decides: owner, after the per-
video baseline)*
85 2. **Per-video target** ? "at-par-with-Gemini-vs-golden on every clip," or an absolute
F1 floor? *(decides: owner)*
86 3. **Recall lever** ? detector-sensitivity tuning vs ensemble fusion as the primary path
for G1. *(decides: data, post-P3)*
87
88 ## 9. Definition of done
89 - [ ] Golden corpus covers the Boxhill set + enough variety that per-video reads are
trustworthy (n meaningfully > 12).
90 - [ ] The **rally-END quick-return** failure mode (G3) is closed: measured ? merge-rate
+ ? |?end| on the clips that exhibit it, no regression elsewhere.
91 - [ ] Local is **at-par-or-better than Gemini vs golden on every corpus video** (no per-
video F1 regression).
92 - [ ] The winning levers are flipped **default-ON** via a Launch Report (over-seg guard
clean on both rulers).
93 - [ ] ? flip `Status ? DONE`, archive to `docs/archives/`.
94
95 ## 10. References
96 **Code anchors `[verified]`: backend/pipeline/detectors/rally_gate.py (Gate A) ·
backend/pipeline/detectors/tracknet_runner.py (R4 end-trim, `_trim_inactive_tail`) ·
backend/eval/windowing.py · backend/eval/rally_seg_eval.py + tolerance_metrics.py (R2) ·
backend/pipeline/segmenters/yolo_hybrid.py (person cue) · scratch/ingest_golden.py,
scratch/boxhill17_score_golden.py.
97 **Internal docs:** [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md),
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) §4,
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md),
[EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), [GOLDEN_VIDEOS.md](GOLDEN_VIDEOS.md),
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
98
99 ---
100
101 ### Changelog
102 - 2026-06-18 ? Created. Seeded with the GX010141 golden scoring (first ground-truth
clip), the three named gaps (recall / precision / rally-END), and the player-state end-rule
design from the owner's quick-return insight.
103 - 2026-06-18 ? Added GX010137 (golden #13, normal match play): local F1@0.5 **0.82**
(recall **0.88 > Gemini 0.79**) vs Gemini 0.89. Confirms **opposite failure modes** (local high-
recall/low-precision, Gemini the reverse) ? **precision is local's #1 lever**, fusion the
durable win. Flagged the GX010141 pre-annotator-fix data caveat.
104 - 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF),
**measured n=15.** Eval `--min-duration`/`--stratify` (rally_seg_eval.py) + product
stitching.min_rally_duration. Finding: single-court over-fire is short-FP-dominated (seg_ratio
1.43?1.04) but **multicourt over-fires 1.58x even on long rallies** (? needs court-localization,
Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 +
added §7 row P7.
105

```

14. assistant (2026-06-19T15:47:11.344Z)

Now I have a clear picture of RALLY_DETECTION_GAP_CLOSURE.md. It's interesting ? this doc has internal inconsistencies: §0 TL;DR and §6 say "n=13"/"n=12", §3 says "n=13"/"n=15", §9 says "n > 12". Several of these are present-tense CURRENT corpus claims that contradict n=15. But I must be careful: some are clearly historical (n=2 ground-truth-scored findings tied to a date). Let me carefully analyze and check the remaining docs.

Let me check the other docs in my slice.

15. user (2026-06-19T15:47:12.809Z)

```
1      # Rally-Detection Quality: Research, Eval Framework & Project Plan
2
3      **Status:** Foundation doc (2026-06-16), run as a **tracked project** (§7) ? **Bridge
the
4      local?Gemini rally-detection gap,** **completed when all committed tiers land and a
single quality
5      number is attributed (§7.1).** The empirical + research backbone for closing the gap
between the
6      **local, no-cloud rally segmenter** and the **Gemini** path. Iterations land as their
own measured
7      PRs, logged in [ `QUALITY_ITERATIONS.md` ](QUALITY_ITERATIONS.md); this doc is the plan
they execute against.
8
9      **How to read this:** §1 frames the gap with measured numbers. §2 is the precise, code-
grounded
10     diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we
already
11     have. §4 specifies how we strengthen that harness (the "really strong evaluation
framework").
12     §5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized
roadmap with
13     falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing,
the single
14     number we attribute to the effort, and the definition of done.** §8 recaps the non-
negotiable
15     constraints + cross-links.
16
17     > **This doc does NOT duplicate prior design.** It builds on, and cross-links:
18     > [ `HOW_RALLY_DETECTION_WORKS.md` ](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental
model),
19     > [ `EXPERIMENT_HARNESS.md` ](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
20     > [ `MULTI_SIGNAL_FUSION_PLAN.md` ](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio
fusion),
21     > [ `ANALYZERS_AND_RECONCILERS.md` ](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion
framework +
22     > §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS
+
23     > late-fusion; we *cite, don't re-derive*), [ `ABLATION_LAYER.md` ](ABLATION_LAYER.md),
24     > [ `EVALUATION.md` ](EVALUATION.md),
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
25     > (TAL-head-first ? VLM), and [ `GOLDEN_SET_VENDORING.md` ](GOLDEN_SET_VENDORING.md)
(corpus growth).
26
27     ---
28
29     ## 1. The gap (measured)
30
31     Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video
semantically)
32     is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud)
lags
33     badly. Measured F1 vs golden labels ([ `QUALITY_ITERATIONS.md` ](QUALITY_ITERATIONS.md)):
34
35     | Path | Mahadevpura (clean, single court) | Kushagra (multi-court) |
36     |---|---|---|
37     | Heuristic velocity windowing (local today) | 0.657 | **0.157** |
38     | Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6) | 0.744 | 0.561 |
39     | Two-stage WASB + Gemini refine | 0.83 | ? |
40     | **Gemini wrapper (gemini-flash)** | **0.93** | **0.66** |
41
42     The local heuristic is **footage-fragile**: fine on a clean single court, collapses on
```

```

43 multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court
gap
44 (0.561 vs 0.66) ? the residual is corpus-sized, not architecture-sized.
45
46 ### 1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)
47
48 A fully-local run (`wasb_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches
49 (Kushagra/Yash June-12; report: `~/June 12 Local-vs-Gemini/_LOCAL_vs_GEMINI_REPORT.md`,
50 generator `scratch/compare_local_vs_gemini.py`). Gemini is a strong reference here,
NOT
51 ground truth (these clips are unlabeled) ? so this is an agreement/divergence study:
52
53 | | Gemini | Local (WASB no-AI) |
54 |---|---|---|
55 | rallies / windows (3 videos) | 59 | 12 |
56 | total "play" | 6:28 | 17:41 |
57 | mean window duration | 6.1 s | 65 s (10.6×) |
58 | true misses (gemini-only) | ? | 0 |
59 | merge groups (one local window ? 2 Gemini rallies) | ? | 7 |
60
61 Headline: the local path does not miss rallies ? it fails to segment them. One
62 mahadevpura-1 window spanned the entire 174 s clip. Local "play" is 2.7× Gemini's
because its
63 mega-windows swallow the dead time between rallies.
64
65 ---
66
67 ## 2. Diagnosis (code-grounded ? read before proposing fixes)
68
69 The local pipeline is two layers (see
[ `HOW_RALLY_DETECTION_WORKS.md` ](HOW_RALLY_DETECTION_WORKS.md)):
70 (L1) detection ? per-frame shuttle position (WASB/TrackNet); and (L2)
segmentation ?
71 turn the trajectory into rally intervals. The gap is almost entirely L2.
72
73 ### 2.1 L1 (shuttle tracking) is NOT the bottleneck
74 On the divergence clips WASB reported the shuttle visible in 96?99 % of frames
(mahadevpura-1
75 98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory signal is present and
dense.
76
77 ### 2.2 L2 (segmentation) is the bottleneck ? and the flaw is conceptual
78 The windowing is `trajectory_to_action_windows`
79 ([`backend/pipeline/detectors/tracknet_runner.py:256`](../backend/pipeline/detectors/tra
cknet_runner.py)):
80
81 - A frame is "active" iff the shuttle is visible AND its per-second-normalized
speed
82 exceeds `velocity_threshold` (default 0.02). Velocity = `(dist / frame_width) * (fps /`
dframes)`
83 ? already fps-normalized (? fraction of frame-width per second).
84 - Active frames within `dead_time_merge_gap` (2.0 s) are merged into one window.
85 - Windows shorter than `min_action_window_duration` (2.0 s) are dropped.
86
87 The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress."
During
88 dead time the shuttle is still visible and gently moving (being picked up, walked back,
knock-ups,
89 slow serve prep) ? those frames read as active. The 2 s merge then bridges the brief
gaps, and
90 because there is no `max_window_duration` cap and no inactivity/boundary split, a
single
91 window grows until the shuttle finally goes invisible ? which, at 96?99 % visibility, is
rarely.
92 Result: rally + dead-time + rally + ? collapse into a few mega-windows.

```

```

93
94     > **Correction to an earlier hypothesis (honesty note).** A first pass guessed the cause
was
95     > "velocity threshold not fps-invariant." It IS fps-invariant (the `fps/dframes`
factor). The
96     > real cause is the *motion-equals-rally* conflation + 2 s bridging + no upper bound.
fps matters
97     > only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above
the
98     > speed threshold. The `_LOCAL_vs_GEMINI_REPORT.md` has been corrected to match.
99
100    ### 2.3 What this implies
101    The local path **has no rally-STATE signal** ? only "is the shuttle moving." Bridging
the gap is
102    fundamentally about giving L2 signals that separate *rally* from *shuttle-merely-moving*
103    (net-crossings, two-sided play, sustained fast exchange, hit sounds) and a segmenter
that
104    produces *bounded, well-cut* intervals ? exactly the multi-signal-fusion + learned-
segmenter
105    direction the existing docs lay out. This doc makes that concrete and measurable.
106
107    ---
108
109    ## 3. What we already have (eval/experiment harness inventory)
110
111    The harness is rich and composable ? we **extend** it, not reinvent it. (Modules under
112    `backend/eval/`.)
113
114    **Scoring & matching**
115    - `metrics.py` ? temporal-IoU interval matching; precision/recall/F1, mean_iou, boundary
errors.
116    - `segmentation_metrics.py` ? **F1@{0.1,0.25,0.5}**, **mAP@tIoU** (0.3?0.7),
**segment_count_ratio**
117    (the over/under-segmentation diagnostic). (Edit/segmental score deliberately omitted
to date.)
118
119    **Statistical rigor (small-n)**
120    - `eval_stats.py` ? bootstrap 95 % CIs, paired exact **sign-test**, `paired_delta_ci`,
121    `grouped_folds` (LOGO), `out_of_fold_predictions` (anti-stacking-leakage). The
promotion bar =
122    "F1 CI excludes 0 **AND** sign-test agrees."
123    - `score_calibration.py` ? Platt + isotonic calibrators, **Brier + ECE**.
124
125    **Variants, A/B, leaderboard**
126    - `experiment.py` ? `Variant` (declarative recipe: segmenter + config + cue_flags +
gates +
127    learned-classifier + threshold/merge_gap/calibrate), `compare()` (LOVO-honest A/B with
an
128    "honest" stats block), `compare_unlabeled()` (agreement on unlabeled footage),
`CONTROL`
129    (pinned baseline), `record_experiment()` ?
`eval_baselines/experiment_leaderboard.json`.
130    - `ablation.py` ? leave-one-signal-out marginal F1 with CI + sign-test + a no-over-
claim verdict
131    vocabulary (`earns_keep` / `suggestive` / `inert` / `structural_protected`).
132    - `classifier.py` ? dependency-free numpy `LogisticRegression` (the learned scorer;
JSON-serializable).
133
134    **Eval?serve discipline + gates**
135    - `windowing.py` ? single-sources the **SERVED** preset from `IndexingConfig` defaults
so eval
136    and production can't silently diverge; `PROPOSAL` (recall-oriented) is where the
golden feature
137    substrate is built. (This exists *because* Gate-A measured **+0.074 on PROPOSAL but
?0.008 on

```

```

138     SERVED** and was reverted ? the canonical eval?serve lesson.)
139 - `promotion.py` / `served_gate_a.py` ? promote a cue only if it does NOT regress at the
served
140     windowing; `per_user_gate.py` / `per_user_regression.py` ? per-user analogues;
`regression.py` ?
141     golden-set no-regression baseline with a GT-hash to detect label changes.
142
143     **Golden substrate**
144 - `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` /
145     `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
146     (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-
detecting.
147 - **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected
via the
148     Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).
149
150     **Signals available to L2 (built, default-OFF):** net-crossing count (`rally_gate.py`,
Gate-A),
151     5 person/court features (`person_detector.py` + `fusion_features.py`), 3 audio-hit
features
152     (`audio_features.py`). Fusion hooks: `fusion_hybrid.py::_enrich_candidate_stats` /
153     `_select_for_handoff`. Persisted in `candidate_segments.stats`.
154
155     ### 3.1 Known gaps in the harness (what §4 fills)
156     1. **Over-segmentation is under-surfaced in the default report.** `segment_count_ratio`
exists but
157         the merge/split *breakdown* (the very thing that exposed the over-merge) lives only
in a scratch
158         script. tIoU-F1 alone is **blind to over-merge** ? a giant window can still "match"
via overlap.
159     2. **No segmental edit-score** (Levenshtein over the predicted/gt label sequence) ? the
standard
160         TAS over-segmentation metric.
161     3. **Per-cue calibration is fitted but never *measured* post-hoc** (no ECE/Brier delta
reported).
162     4. **`out_of_fold_predictions` (anti-leakage stacking) is built but not wired to an
arbiter.**
163     5. **No active-learning loop** ? nothing tells us *which* unlabeled video to label next.
164     6. **No single "rally-segmentation eval" entrypoint** ? the pieces exist but a one-
command
165         `score this segmenter vs golden, honestly` runner would make every iteration cheap.
166     7. **The local-vs-Gemini agreement track is ad-hoc** (a scratch script), not a first-
class
167         *shadow* signal alongside the golden gate.
168
169     ---
170
171     ## 4. The strengthened evaluation framework (the "really strong harness")
172
173     Design principles (inherited, non-negotiable):
174     - **Signals ? learned scorer, NO special-casing**
([`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) §1):
175         every new cue is a feature column the scorer learns to weight; never an `if/else` in
the decision path.
176     - **Promote only on measured lift; default OFF; revert is one flag**
([`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) §6).
177     - **eval ? serve:** a harness win must be re-confirmed on the *served* windowing before
any default flip.
178     - **Honest small-n stats:** never a bare mean at n?6?16; always CI + paired sign-test
(C1).
179
180     ### 4.1 Metric set the leaderboard should standardize on
181     Surface ALL of these in the default `compare()` report (most already computed; the task
is to
182     *report* them together so over-merge can never hide again):

```

```

183
184 - **tIoU-F1 @0.5** (primary) **and F1@[0.1,0.25,0.5]** (segmental, TAS-standard).
185 - **mAP@tIoU** (0.3?0.7) when predictions carry confidences.
186 - **`segment_count_ratio`** + a **merge-rate / split-rate** breakdown (fraction of GT
rallies that
187 share a predicted window = under-segmentation; fraction of predicted windows that span
?2 GT =
188 the over-merge signal). Promote the scratch `compare_local_vs_gemini.py` matching into
a reusable
189 `backend/eval/` helper.
190 - **Segmental edit (Levenshtein) score** ? the canonical over-segmentation metric we
currently lack.
191 - **Boundary timing** (mean start/end error on matched pairs) ? already in `metrics.py`,
surface it.
192 - **Calibration (ECE/Brier)** reported pre/post per-cue calibration (close gap #3).
193
194 > **Why:** plain tIoU-F1 rewarded our mega-windows (they "overlap" real rallies). The
metric set
195 > above makes over/under-segmentation *first-class*, so a fix that improves segmentation
is
196 > visible even when tIoU-F1 barely moves ? and vice-versa.
197
198 ### 4.2 Splits & significance
199 - **Leave-one-video-out, grouped by match** (never leave-one-*window*-out ? windows
within a video
200 are correlated). Keep `grouped_folds`. Note the **RL00CV bias** caveat for tiny n (see
$5.4).
201 - **Bootstrap CI + paired exact sign-test** on per-video held-out F1 (C1). A result is
*promotable*
202 only if CI excludes 0 **and** the sign-test agrees, **and** it survives the served-
windowing check.
203 - **Anti-leakage:** per-LOVO-fold threshold tuning + calibration on the TRAIN fold only
(already
204 enforced in `experiment._calibrate_and_tune`); wire `out_of_fold_predictions` once an
arbiter exists.
205
206 ### 4.3 The golden corpus + active learning (the highest-leverage lever)
207 The single biggest quality lever is **more labeled matches**
([`NEXT_STEPS.md`](NEXT_STEPS.md):
208 "corpus growth = highest leverage"). With ~6 ? ~16 videos:
209 - Grow via the Roora vendoring flow
([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md),
210 [`ROORA_LABELING_GUIDELINES.md`](ROORA_LABELING_GUIDELINES.md)) ? court calibration,
IAA, the
211 3 non-negotiables (ignore dead time, every rally, one-contact = one shot).
212 - **Active learning:** stop labeling videos at random. Score the *unlabeled* corpus and
label the
213 videos where the local segmenter is most *uncertain* / most *divergent from Gemini* /
most
214 *diverse* (new venue, fps, lighting). $5.3 surveys the methods; the local-vs-Gemini
agreement
215 signal ($4.4) is a ready proxy for "where do we disagree most ? label that."
216
217 ### 4.4 Shadow evaluation: Gemini-agreement as a free, unlimited signal (NOT a gate, NOT
training data)
218 We have exactly one strong reference available on *unlabeled* footage at inference time:
**Gemini**.
219 - `experiment.compare_unlabeled()` already scores agreement (agreed / a-only / b-only)
without labels.
220 - Use it as a **shadow metric**: run any candidate segmenter on a large unlabeled pool,
measure
221 agreement-with-Gemini, and watch it as a *trend* ? cheap, unlimited, catches
regressions the 6
222 golden videos can't. **It is a shadow signal only.** Golden labels remain the gate;
Gemini is not

```


223 ground truth (it has its own errors) and ? per the licensing guardrail (§8) ? **its**
 outputs may
 224 never become training data for owned weights.**
 225
 226 **### 4.5 One-command rally-segmentation eval (close gap #6)**
 227 Wrap the above into a single reusable entrypoint (a future PR ? not built here): given a
 segmenter
 228 variant + the golden manifest, emit the full §4.1 metric set + the honest stats block +
 a leaderboard
 229 row, and (optionally) the §4.4 shadow agreement on an unlabeled pool. Every iteration in
 §6 is then
 230 "run the entrypoint, read the honest delta, promote-or-not."
 231
 232 ---
 233
 234 **## 5. External research synthesis (cited, adversarially verified)**
 235
 236 > Produced by a deep-research run (6 search angles ? 27 sources ? 126 candidate claims ?
 25
 237 > 3-vote-adversarially-verified, 23 confirmed / 2 refuted). ****Cite, don't over-claim.****
 Two claims
 238 > were ***killed*** in verification ? recorded in §5.5 so we don't repeat them. ****License**
caveat:** the
 239 > code repos below are research releases; ****independently confirm each repo's actual**
 license is
 240 > MIT/Apache-2.0/BSD before vendoring any of it****** (this survey did not verify every
 license file).
 241 > ****Governance:**** none of these techniques require training on paid-LLM outputs ? all
 are compatible
 242 > with our no-Gemini-training-data guardrail (§8).
 243
 244 **### 5.1 Windowing & boundaries (TAL/TAS) ? the explicit-boundary machinery we lack**
 245 The recurring lesson: modern action-localization models have an ****explicit boundary**
mechanism**;
 246 our heuristic has none (it merges on a velocity+gap rule). The transferable ***idea*** is
 "predict a
 247 boundary and cut there," but the deep models are built for ****hundreds+** of training
 videos******, so at
 248 n=6?16 they are ****research bets****, not drop-ins.
 249
 250 - ****ASRF ? Action Segment Refinement Framework**** (Ishikawa et al., WACV 2021,
 [arXiv:2007.06866](https://arxiv.org/abs/2007.06866)).
 251 A shared feature extractor feeds ****two branches****: frame-wise classification + a
****Boundary**
 252 Regression Branch******; predicted boundaries refine/split the segmentation (+13.7% edit,
 +16.1%
 253 segmental F1 over MS-TCN-era SOTA on 50Salads/GTEA/Breakfast). ****Why it fits:**** the
 boundary head
 254 is exactly the missing "cut here" mechanism. ****? Two caveats:**** (1) ASRF's headline
 goal is to
 255 ***reduce*** over-segmentation (merge spurious fragments) ? the opposite of our
 over-***merge***; only the
 256 ***split-at-a-detected-boundary*** sub-mechanism transfers. (2) The claim that ASRF's
 boundary branch
 257 is a ****model-independent post-hoc splitter**** you can bolt onto any dense signal was
****REFUTED****
 258 (§5.5) ? you must train the two-branch model, not graft the branch. ****Effort/payoff:****
 research bet.
 259
 260 - ****TriDet**** (Shi et al., CVPR 2023,
 [arXiv:2303.07347](https://arxiv.org/abs/2303.07347)). A
 261 ****Trident-head**** models each boundary as a ***relative probability distribution over**
 local bins*
 262 (boundary = expected value over neighboring instants) ? built for imprecise/ambiguous
 boundaries.

263 69.3% avg mAP on THUMOS14 (> ActionFormer 66.8% at ~75% latency), **convolution-only**
 (SGP), no
 264 self-attention (attractive on a single local GPU). Ablation shows the Trident-head's
 gain
 265 concentrates at strict tIoU=0.7 ? it buys **boundary precision**. **Effort/payoff**:
 research bet
 266 (trained on 100s of videos); the portable piece is the relative-boundary idea.
 267
 268 - **ActionFormer** (ECCV 2022, [arXiv:2202.07925](https://arxiv.org/abs/2202.07925)).
 The reference
 269 single-stage anchor-free template: multiscale pyramid + local self-attention; classify
 every
 270 moment + regress its boundaries (71.0% mAP@0.5 THUMOS14). This is the family
 TriDet/TemporalMaxer/
 271 CausalTAD extend, and the natural target if we ever graduate from heuristic+LogReg.
Research bet.
 272
 273 - **TemporalMaxer** (CVPRW 2023, [arXiv:2303.09055](https://arxiv.org/abs/2303.09055)) ?
the most
 274 important small-data signal in this dimension. It replaces ActionFormer's self-
 attention block
 275 with a **parameter-free max-pool** and **beats** it (67.7 vs 66.8 avg mAP) at **~4x**
fewer params.
 276 The ablation is the takeaway: a **learned 1-D conv block** is the **WORST** (59.4,
 overfits redundant
 277 features); parameter-free pooling wins. **Why it fits**: direct evidence to **keep our**
 scorer
 278 parameter-light (the numpy LogisticRegression over a handful of features is the
right choice at
 279 our scale, not a placeholder to rush past). **Effort/payoff**: a discipline (free) ?
 adopt the
 280 principle, not the model. (Caveat: the overfitting evidence is about feature
 redundancy on a
 281 mid-size set, a directional inference to our small-n setting.)*
 282
 283 - **CausalTAD** (2024, [arXiv:2407.17792](https://arxiv.org/abs/2407.17792)) ?
steering-away note.
 284 Causal-attention masking (attend only to past/present) won EPIC-Kitchens/Ego4D 2024
 tracks, but it
 285 adds **no** boundary-split module. Useful later for **online/streaming** rally
 detection; **NOT** the
 286 over-merge fix. Listed so we don't mistake causal attention for the remedy.
 287
 288 - **Metrics** (the actionable cross-cut): the field leads with **segmental Edit**
 (Levenshtein) score
 289 and **F1@k**, not frame accuracy (MoF) or plain tIoU ? see §5.4.
 290
 291 **### 5.2 Shuttle-trajectory ? rally signal processing ? published badminton systems**
 already do what we lack
 292 This is the highest-confidence, lowest-effort dimension: in-domain papers operationalize
 the exact
 293 boundary heuristics we're missing, in spirit license-compatible (re-confirm before
 vourcing code).
 294
 295 - **See et al. ? Service & End-of-Rally Detection in Badminton** (2024 Springer chapter
 296
 [10.1007/978-3-031-69073-0_15](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15);
 297 2025 SN Computer Science follow-on
 [10.1007/s42979-025-03935-0](https://link.springer.com/article/10.1007/s42979-025-03935-0)).
 298 **(a) Rally END = a sustained-inactivity rule**: the rally ends when **~80%** of
 consecutive frames
 299 show no shuttle motion (**~95% accuracy** over 187 rallies). **(b) Rally START**
 (service) = a
 300 learned classifier over court geometry + player poses (~26 features/frame × 12
 frames ?

301 CNN/BiLSTM, ****88.6%**** accuracy). ****Why it fits:**** the inactivity rule is ***precisely***
 the
 302 end-of-rally cut our windowing lacks ? a few lines, directly breaks 65 s mega-windows.
****? Caveats:****
 303 the rule is a ***termination***, not a ***within-rally split***; results rest on paywalled
 abstracts +
 304 small test sets (35?187 rallies); the serve classifier needs the (default-OFF) person
 cue + court
 305 polygons. ****Effort/payoff:**** the inactivity rule = ****low-effort high-confidence win****;
 serve
 306 classifier = medium.
 307
 308 - ****Hsu, Yu & Cheng ? trajectory + swing fusion**** (Sensors 2024, 24(13):4372,
 309 [10.3390/s24134372](https://www.mdpi.com/1424-8220/24/13/4372)). Fusing TrackNet
 trajectory with a
 310 YOLOv7 swing detector via a Shot Refinement Algorithm raises hit detection ****F1 72.3%**
? 90.5%**
 311 (over 69 BWF matches / 1582 rallies). Hits are found by ****trajectory direction-**
 reversal / peak
 312 detection** (convert (x,y)?(frame,y) wave peaks) then reconciled with the swing cue.
****Why it fits:****
 313 validates our ****late/score-level fusion**** approach in-domain, and direction-
 reversal/peak is a cheap
 314 ****per-window feature**** marking contact events (a "rally has hit cadence" signal ? the
 discriminator
 315 a "shuttle-moving" heuristic lacks). ****? Important caveat (the one 2-1 vote):**** this
 paper does
 316 ***within-rally stroke/hit* segmentation, **not** rally start/end ? "it splits merged**
 rallies" is ***our**
 317 extrapolation*; frame it as "hit cadence plausibly helps locate rally boundaries," not
 a proven
 318 rally-splitter. ****Effort/payoff:**** the peak/direction-reversal feature = low-effort
 win (feed the
 319 scorer); full swing fusion = medium.
 320
 321 - ****What best separates "rally" from "shuttle merely moving"***** (synthesis): sustained-
 inactivity
 322 termination (See), hit cadence / direction-reversals (Hsu), net-crossings ?2 (our
 Gate-A,
 323 [rally_gate.py](../backend/pipeline/detectors/rally_gate.py)), two-sided court
 occupancy (our
 324 person features). None alone is sufficient; the spine (\$4) says ****feed them all to the**
 learned
 325 scorer**.
 326
 327 **### 5.3 Small-data strategies ? most quality per label**
 328 - ****Timestamp / sparse supervision**** (Li et al., CVPR 2021,
 [arXiv:2103.06669](https://arxiv.org/abs/2103.06669);
 329 TAS survey Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352)). Label
****one frame per**
 330 action** ? iteratively refine pseudo frame-labels ? ***comparable*** to full supervision
 (50Salads:
 331 timestamp 75.6% acc / 60.1 F1@50 vs full 83.7% / 70.1). ****Why it fits:**** sparse per-
 rally timestamps
 332 cost far less than dense boundary labels ? relevant if we ever train a frame-wise
 model.
 333 ****? Caveats:**** "comparable" hides a real ****~8?10 pt residual gap****; annotation savings
 are modest
 334 (~35%, annotators still watch the whole clip); benchmarks are cooking/egocentric ?
 transfer to 1-D
 335 shuttle trajectories is ****unproven****. ****Effort/payoff:**** research bet to pilot.
 336 - ****Parameter-light scorer**** ? see TemporalMaxer (\$5.1): keep the numpy
 LogisticRegression; resist a
 337 learned temporal net until the corpus is much larger. ****Low-effort win (a**
 discipline).******

338 - **? OPEN (the research could NOT verify these ? honest gap):** *active learning*
(which video/segment
339 to label next via uncertainty/diversity/core-set) and the *teacher-student /
distillation governance
340 boundary* for using Gemini as a weak/pseudo-labeler returned **no surviving verified
claim**. Our
341 pragmatic, guardrail-safe stance (not from the literature ? from our own design): pick
the next
342 video to label by **maximal local-vs-Gemini disagreement** (the §4.4 shadow signal) +
**scorer
343 uncertainty** + **venue/fps diversity**; and treat **Gemini strictly as an inference-
time
344 weak-reference / eval oracle ? never a training-data source for owned weights** (§8).
These two
345 topics deserve their own grounded research pass before we lean on them.
346
347 ### 5.4 Evaluation rigor at small n ? lead with segmental metrics
348 - **Frame accuracy (MoF) and plain tIoU are BLIND to over-segmentation** (TAS survey,
Ding et al.,
349 [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352); segmental-F1 origin Lea et al.,
CVPR 2017).
350 "The score can be high even when segments are fragmented." The over-merge that bit us
is exactly
351 what tIoU-F1 hides (a giant window still *overlaps* real rallies). **Action:** lead
the leaderboard
352 with the **segmental Edit (Levenshtein) score** (penalizes spurious
ordering/fragmentation) **and
353 F1@{0.1,0.25,0.5}**, paired with our existing `segment\_count\_ratio` + the merge/split
breakdown.
354 **We already have** F1@k, mAP@tIoU, segment-count-ratio, bootstrap CIs, paired sign-
test, Platt/
355 isotonic + ECE/Brier ? the only *missing* primary metric is the **Edit score** (low-
effort add).
356 - **? OPEN (not covered by a verified claim):** IAA / label-noise level in the golden
set, and which
357 over-segmentation-robust loss (ASRF boundary/smoothing, MS-TCN truncated-MSE) best
tolerates it at
358 small n. Measure IAA as the corpus grows (Roora has an A/A overlap plan); revisit
robust losses
359 only when a frame-wise model is on the table.
360
361 ### 5.5 Refuted claims (recorded so we do NOT repeat them)
362 1. **"ASRF's boundary branch is model-independent ? a post-hoc splitter applicable to
any dense
363 segmentation output."** **REFUTED (0-3).** Implication: we cannot bolt ASRF's
boundary head onto
364 our trajectory signal as a drop-in; the two-branch model must be trained together. ?
port the
365 *concept*, don't expect a plug-in.
366 2. **"A GFL-inspired Boundary Distribution Regression head is the mechanism most
relevant to splitting
367 merged windows."** **REFUTED (0-3)** (and the cited source had a suspect identifier).
Do not adopt
368 this framing as established.
369
370 ### 5.6 External due-diligence review (2026-06-17) ? owner-commissioned; verify before
submission
371 An owner-commissioned external review (31 cited sources) of the
372 RALLY_DETECTION_QUALITY_REPORT.md checkpoint
**validated the
373 methodology** (the decoupled 2-layer design, the R2 metric, LOVO + paired-sign-test
discipline, the
374 consented-data + over-seg-aware-harness moat) and surfaced these **borrowable inputs**.
These are
375 *research inputs for later pickup* ? they add **no measured numbers**, and the citations

are the

376 review's own and ****must be re-verified before any submission****.

377

378 - ****Positioning ? the "first-mile" gap (highest-value).**** Badminton stroke-classification SOTA

379 (****BST**** ? Badminton Stroke-type Transformer, CVPR-W 2026; ****DSTA**** ? Decoupling Spatio-Temporal

380 Adapter) and the big datasets (****ShuttleSet**** ~36,492 strokes; ****Fine-Badminton**** ~27,597 actions)

381 all ****silently assume a pre-trimmed rally clip already exists**** ? they depend on a human having

382 segmented the untrimmed match. Our system solves that ignored ***prerequisite***. Reframe the paper as

383 *****"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage"***** ?

384 a real, unoccupied gap and a far stronger narrative than "applied/systems."

385 - ****R2 ? Action-Spotting / Precise-Event-Spotting (AS/PES).**** Our asymmetric tolerance-match metric

386 sits at the ****TAL × AS intersection**** (interval task, tolerance-based match). Grounding it in the

387 AS/PES literature (SoccerNet, TenniSet, T-DEED, the PES survey [arXiv:2505.03991]) frames R2 as a

388 deliberate methodological contribution, not an ad-hoc metric ? strengthens §5.4 and the report's §9.

389 - ****Comparative baselines reviewers will require.**** Benchmark against ****MonoTrack**** and ****BST**** on

390 ****ShuttleSet / Fine-Badminton**** ? we currently have only a generic heuristic + the Gemini wrapper;

391 domain-specific baselines are a hard publication requirement.

392 - ****Scorer upgrade ? prefer TemporalMaxer.**** For the post-logreg scorer (corpus-gated; NEXT_STEPS M0.4),

393 ****TemporalMaxer**** (parameter-free temporal max-pool; ~2.8× fewer GMACs; robust at small-N,

394 [arXiv:2303.09055]) fits the cheap/local/fast mandate better than attention-heavy ActionFormer.

395 - ****Court polygons ? pose ? the rally-END lever.**** Court masking is not only the person-cue spectator

396 fix (§4 / `MULTI\_SIGNAL\_FUSION\_PLAN.md`): once pose is isolated to on-court players it can resolve

397 ****ambiguous, occluded end-of-rally states**** ? feeding directly into the ****#1 remaining gap (the**

398 rally-END boundary)** , not just precision. Connect court-polygons to the R4/END-trim work.

399 - ****Name the eval?serve lesson "pipeline distribution skew."***** Downstream-cue value depends on the

400 upstream proposal distribution (the Gate-A episode) ? a citable framing and candidate paper "lesson."

401 - ****Citable Layer-1 anchor (public benchmark, NOT our footage).**** WASB SBDT F1 95.6 (step=1) / 94.0

402 (step=3) vs MonoTrack 92.1, TrackNetV2 89.4, BallSeg 71.7 (WASB, BMVC 2023). Justifies the frozen

403 detector choice; ****keep distinct from our own, still-unmeasured footage domain-gap**** (§2.1, a real

404 open limitation the review also flags).

405

406 ****Sequencing:**** these are ***publication-prep + post-current-idea*** levers. The prerequisites remain

407 corpus growth to n?16 and ****multi-court isolated as its own reported slice**** (§6 Tier-3 / §7) ? do

408 those first; the items above land once current windowing/fusion ideas are exhausted.

409

410 ---

411

412 **## 6. Prioritized roadmap (cheap ? durable)**

413

```

414 Every item is its own measured PR: default-OFF, scored on the §4 metric set over the
golden set
415 with honest stats (CI + sign-test), re-confirmed on the served windowing (the
eval?serve gate),
416 logged in [ `QUALITY_ITERATIONS.md` ](QUALITY_ITERATIONS.md). Ordered by the research's
417 effort/confidence tiering. Each carries a falsifiable success criterion ? if it
isn't met, the
418 item does not promote (and that null is itself recorded).
419
420 Definition of "bridging the gap"
421 Local rally segmentation, measured honestly LOVO-by-match on the golden set,
approaches the
422 Gemini reference (today 0.93 single-court / 0.66 multi-court tIoU-F1) ? but tracked
primarily
423 through the over-segmentation-sensitive metrics (segmental Edit + F1@k + low
merge/split rate),
424 since those, not bare tIoU-F1, capture our actual failure. A secondary, free signal is
rising
425 agreement-with-Gemini on the unlabeled pool (§4.4, shadow only).
426
427 Tier 0 ? Make the failure measurable FIRST (do before any fix)
428 - E1 · Over-segmentation-aware leaderboard. Add the segmental Edit (Levenshtein)
score to
429 `segmentation_metrics.py`; surface Edit + F1@{0.1,0.25,0.5} + `segment_count_ratio` +
a
430 merge-rate / split-rate breakdown (promote the
`scratch/compare_local_vs_gemini.py` matching
431 into a reusable `backend/eval/` helper) in the default `experiment.compare()` report;
add the
432 one-command rally-segmentation eval endpoint (§4.5). Success: the current
heuristic's
433 over-merge shows up as a number (low Edit, high merge-rate) on the 6 golden videos,
and a
434 baseline row lands in the leaderboard. (Low effort, high confidence ? research-backed
§5.4.)*
435
436 Tier 1 ? Low-effort, high-confidence wins
437 - W1 · Bounded windowing (the direct over-merge fix). Add a `max_window_duration`
cap and a
438 sustained-inactivity split to `trajectory_to_action_windows` ? port See et al.'s
rule (split /
439 end a window when ~80% of consecutive frames over a trailing window show no shuttle
motion).
440 Config-gated, default preserves current behavior. Success: segmental Edit + F1@k
improve vs
441 CONTROL on the golden set (?F1 CI excludes 0 and sign-test agrees) with no
served-windowing
442 regression; merge-rate drops materially. (Win ? §5.2 See et al., ~95% end-of-rally
acc.)*
443 - W2 · Rally-state features into the learned scorer (no special-casing). Wire the
already-built,
444 default-OFF cues as scorer feature columns: net-crossings (Gate-A), two-sided
motion, and a
445 new cheap trajectory feature ? direction-reversal / peak count (Hsu) as a hit-
cadence proxy;
446 audio hit-rate when the hit detector lands. Re-measure with W1 in place and on the
served
447 windowing ? recall person was ?0.021 and audio +0.079 but both CIs spanned 0 at
n=6 on the
448 pre-fix over-merged candidates; the picture may change once windows are bounded.
Success: the
449 ablation (`ablation.py`) shows a cue with ?F1 CI clearing 0 + sign-test, surviving
served re-check
450 ? promote that cue; others stay default-OFF + retained (no deletion on a small-n
null). (Win/medium.)*

```

```

451 - **W3 · Keep the scorer parameter-light (a discipline, documented not coded).** Per
TemporalMaxer's
452 ablation (learned temporal conv overfits; parameter-free wins), resist replacing the
numpy
453 LogisticRegression with a learned temporal net until the corpus is much larger. Record
this as a
454 standing decision. *(Win ? §5.1 TemporalMaxer.)*
455
456 ### Tier 2 ? Medium effort
457 - **M1 · Serve-start classifier ? rally-start boundary** (See et al. pose/court, 88.6%).
Needs the
458 person cue + a **court-mask source** (manual polygon vs auto-homography ? *owner-
gated* per
459 [ `MULTI_SIGNAL_FUSION_PLAN.md` ](MULTI_SIGNAL_FUSION_PLAN.md) §9.1). **Success:**
start-boundary
460 timing error drops on golden; promotes only if it lifts segmental F1@k without served
regression.
461 - **M2 · Per-venue threshold calibration** against golden labels (the heuristic's
footage-fragility ?
462 it scored 0.157 on multi-court). Tune `velocity_threshold` + `merge_gap` so local
segment counts
463 track golden counts per venue. **Success:** multi-court F1 rises toward the Gen-0
head's 0.561
464 without harming clean-court F1. *(Velocity is already fps-normalized ? calibrate per-
venue, not
465 per-fps.)*
466
467 ### Tier 3 ? Research bets (durable; corpus-gated)
468 - **R1 · Owned segmenter with an explicit boundary branch** (the ASRF/TriDet *idea* ? a
boundary head
469 that predicts cut points ? ported into our owned TAL head, NOT the full deep model,
and **trained**,
470 not grafted (§5.5 refutation)). Per
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ](OWNED_MODEL_IMPLEMENTATION_PLAN.md):
471 **TAL head first, VLM later.** **Gate:** the central unverified assumption is whether
boundary-head
472 gains hold at our scale on trajectory features ? so **pilot only when the corpus is
materially
473 larger (~?16?30 matches)** , and only after W1/W2 establish the heuristic ceiling.
**Success:**
474 beats the tuned heuristic + LogReg on held-out matches (segmental Edit + F1@k),
honestly LOVO.
475 - **R2 · Timestamp/sparse supervision** to get more labeled rallies per annotator-hour
(§5.3) ? pilot
476 if/when we train a frame-wise model; research bet (residual gap, unproven on
trajectories).
477
478 ### Throughout ? corpus growth + active learning + label quality
479 - **Grow the golden set 6 ? ~16** via Roora
([ `GOLDEN_SET_VENDORING.md` ](GOLDEN_SET_VENDORING.md)) ?
480 the single highest-leverage lever ([ `NEXT_STEPS.md` ](NEXT_STEPS.md)).
481 - **Active learning:** select the next videos to label by max **local-vs-Gemini
disagreement** (§4.4)
482 + scorer uncertainty + venue/fps diversity (our pragmatic stance; §5.3 flags the
formal methods as
483 an open research pass).
484 - **Measure IAA / label noise** as the corpus grows (Roora A/A overlap); revisit over-
segmentation-
485 robust losses only when a frame-wise model is on the table (§5.4 open question).
486
487 ### Dead-ends / do-NOT (so we don't relitigate)
488 - Don't treat **causal attention** (CausalTAD) as the over-merge fix ? it has no
boundary-split (§5.1).
489 - Don't expect **ASRF's boundary branch as a post-hoc drop-in** on our dense signal ?
refuted (§5.5).

```

```

490 - Don't adopt a **learned temporal net** at n=6?16 ? TemporalMaxer evidence says it
overfits (§5.1).
491 - Don't chase **bare tIoU-F1 / frame accuracy** ? blind to the over-merge (§5.4).
492 - Don't let **Gemini outputs become training data** for owned weights ? inference/shadow
only (§8).
493
494 ---
495
496 ## 7. Project: plan, tracking & definition of done
497
498 This roadmap is run as **one tracked project ? "Bridge the local?Gemini rally-detection
gap."** It
499 **completes when all committed tiers land AND a single quality number is attributed to
the effort**,
500 measured honestly and logged. This §7 is the living burn-down (update the status column
as PRs merge;
501 log each result in [QUALITY_ITERATIONS.md])(QUALITY_ITERATIONS.md), numbers in
502 `eval_baselines/experiment_leaderboard.json`).
503
504 ### 7.1 The number (what we attribute to this effort)
505 Measured **leave-one-video-out, grouped by match, on the golden set** (grown to ?~16):
506 - **Headline:** multi-court rally-segmentation **tIoU-F1**, `baseline ? final`, plus
**gap-closure %**
507 = `(final ? baseline) / (Gemini ? baseline)`. Today: heuristic **0.157** ? Gemini
**0.66**
508 (Gen-0 head already **0.561**). **Success bar (proposed, owner-adjustable): close ?
50% of the
509 heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally
beating Gen-0's 0.561.
510 - **Primary lens (over-segmentation-sensitive, the actual failure):** segmental
**F1@0.25** +
511 **Edit-score** + **merge-rate** ? these must improve even if tIoU-F1 moves little.
512 - **Reported with honest stats** (bootstrap CI + paired sign-test); a result counts only
if it also
513 holds on the **served** windowing (eval?serve gate). The **baseline is frozen at E1**
before any fix.
514 - **Tracked headline = the n=6 golden *aggregate* tIoU-F1@0.5** (LOVO; the robust unit,
not a single
515 video) via `python -m backend.eval.rally_seg_eval`. The 0.157/0.66 figures above are
per-venue
516 **references**; computing an exact "% gap-to-Gemini closed" needs Gemini measured on the
**same* golden
517 set + metric ? **TODO** (don't mix the per-venue refs with the aggregate).
518
519 **Progress (cumulative, golden n=6 aggregate tIoU-F1@0.5):**
520
521 | after | F1 | merge_rate | ? vs baseline |
522 |---|---|---|---|
523 | E1 baseline (current heuristic) | 0.300 | 0.523 | ? |
524 | + W1 bounded windowing (cap=12, **default-OFF**) | **0.439** | **0.038** | **+0.139**
(6/0 sign-test) |
525 | + W2 net-crossings gate (mc=1, **default-OFF**) | **0.457** | 0.038 | **+0.157
cumulative** (W2 step +0.019, 6/0) |
526 | + W1.5 hard-cap fallback (**default-OFF**) | 0.444 | 0.000 | **+0.006 (n.s., CI spans
0)** ? neutral on golden; real-footage fix (mahadevpura-1 174s blob ? 24x?9s windows) |
527 | + W2 multi-feature scorer (person/audio) | _next (GPU)_ | | |
528
529 ### 7.2 Work items (status checklist)
530 Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF,
measured PR.)
531
532 | ID | Deliverable | Tier | Depends on | Acceptance / promote criterion | Status |
533 |---|---|---|---|---|---|
534 | **E1** | merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval`
entrypoint; **froze baseline** (#185) | 0 | ? | over-merge shows up as a number on the 6 golden

```


videos | ? baseline F1 0.300, merge_rate 0.523 |

535 | ****W1**** | Bounded windowing ? `max\_window\_duration` cap + inactivity split (See et al.) (#187) | 1 | E1 | seg F1@k ? vs baseline (CI?0 ****and**** sign-test), no served regression; merge-rate ? | ? ****?F1 +0.139 (6/0), merge_rate 0.523?0.038**** (default-OFF) |

536 | ****W1.5**** | Hard-cap fallback (`window\_hard\_cap`) ? chop continuously-active windows at the cap when no inactivity gap exists | 1 | W1 | golden ?F1 ?0 (or neutral) ******** fixes a real-footage over-merge it can't otherwise cap | ? ****golden ?F1 +0.006 (n.s.)****; real `mahadevpura-1` 174s blob ? 24x?9s; retained ****default-OFF**** (real-footage + true-cap rationale, not a golden lift) |

537 | ****W2**** | Rally-state features (net-crossings ? #188 + ****never-empty safeguard**** ?; two-sided motion, ****direction-reversal/peak****, audio = follow-on) | 1 | E1, W1 | ?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF | ? ****net-crossings ?F1 +0.019 (6/0)****, gate now ****do-no-harm**** (never zeroes a video); multi-feature scorer next (GPU) |

538 | ****W3**** | Keep scorer parameter-light (standing decision; TemporalMaxer evidence) | 1 | ? | documented; no learned temporal net until corpus ? now | ? |

539 | ****M1**** | Serve-start classifier ? rally-start boundary (pose/court) | 2 | W1, ****court-mask decision (owner-gated)**** | start-boundary timing error ?; promote on seg-F1 lift, no served regression | ? |

540 | ****M2**** | Per-venue threshold calibration vs golden | 2 | E1 | multi-court F1 ? toward Gen-0 0.561 without harming clean-court | ? |

541 | ****R1**** | Owned segmenter with an explicit ****boundary head**** (port ASRF/TriDet ***idea***, trained ? not grafted) | 3 (stretch) | corpus ?~16?30, W1/W2 | beats tuned heuristic+LogReg on held-out matches (seg Edit + F1@k), LOVO | ? |

542 | ****R2**** | Timestamp/sparse-supervision pilot (label-efficiency) | 3 (stretch) | R1 path | measured label-efficiency win | ? |

543 | ****C**** | Golden corpus 6 ? ~16 (Roora) + active-learning selection + measure IAA | ongoing | ? | corpus ?16; IAA recorded | ? (~6 now; ~10 by the weekend) |

544

545 **### 7.3 Sequencing**

546 `E1` (metrics + baseline) ? `W1`/`W2`/`W3` (the high-confidence wins) ? `M1`/`M2` ? `R1`/`R2`

547 (corpus-gated). `C` (corpus growth) runs in parallel throughout ? it is the single highest-leverage

548 input and gates the Tier-3 bets.

549

550 **### 7.4 Definition of done**

551 The project is ****complete**** when:

552 1. ****Tier 0 + Tier 1 + Tier 2**** work items are merged ? each a measured PR (default-OFF, promoted only

553 on measured lift, re-confirmed on the served windowing).

554 2. ****The headline number (\$7.1) is recorded**** in `QUALITY\_ITERATIONS.md` + the leaderboard, with honest

555 stats, on the grown golden set ? ****and the gap-closure target is met, or the honest null is recorded**

556 with a reasoned go/no-go on Tier 3.**

557 3. Each landed item left the harness greener (no regression) and `QUALITY\_ITERATIONS.md` tells the story.

558

559 > ****Tier-3 scope decision (owner-adjustable):**** R1/R2 (the owned boundary-head model + sparse

560 > supervision) are ****corpus-gated stretch goals**** ? the verified research says deep TAL detectors risk

561 > overfitting at n=6?16. ****Default:**** the project ***completes*** at Tier 0?2 + an attributed number, and

562 > Tier 3 spins out into the owned-model milestone

563 ([`OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).

564 > Fold Tier 3 into the DoD instead if you want the project to span the owned model.

564

565 ---

566

567 **## 8. Constraints & guardrails (non-negotiable) + cross-links**

568

569 - ****Licensing:**** every dependency ? code, data, AND weights ? must be ****MIT / Apache-2.0 / BSD****.

```
570     Reject viral/NC/custom (Gemma-3, Llama-4 MAU cap, Commons-Clause).
571     - **Never train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs**
(terms/copyright);
572     paid LLMs are inference/serving/shadow-reference only. Also forbidden: public datasets
known to
573     be LLM-generated. **Exclude minors** from any training pool; per-user data needs
explicit opt-in.
574     - **Owned model:** TAL head FIRST (de-risked, trains in minutes, doubles as a pseudo-
label teacher +
575     honest baseline the VLM must beat), then Qwen3-VL (Apache-2.0) / InternVL3 (MIT) ? see
576     [ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ] (OWNED_MODEL_IMPLEMENTATION_PLAN.md).
577     - **Cross-links:** [ `HOW_RALLY_DETECTION_WORKS.md` ] (HOW_RALLY_DETECTION_WORKS.md) .
578     [ `EXPERIMENT_HARNESS.md` ] (EXPERIMENT_HARNESS.md) .
[ `MULTI_SIGNAL_FUSION_PLAN.md` ] (MULTI_SIGNAL_FUSION_PLAN.md) .
579     [ `ANALYZERS_AND_RECONCILERS.md` ] (ANALYZERS_AND_RECONCILERS.md) .
[ `ABLATION_LAYER.md` ] (ABLATION_LAYER.md) .
580     [ `EVALUATION.md` ] (EVALUATION.md) . [ `QUALITY_ITERATIONS.md` ] (QUALITY_ITERATIONS.md) .
581     [ `GOLDEN_SET_VENDORING.md` ] (GOLDEN_SET_VENDORING.md) .
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ] (OWNED_MODEL_IMPLEMENTATION_PLAN.md).
582
```

16. user (2026-06-19T15:47:12.970Z)

```
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\GOLDEN_SET_VENDORING.md
```

17. user (2026-06-19T15:47:13.145Z)

```
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\GOLDEN_SET_VENDORING.md
docs\GOLDEN_DATA_SHARING.md
docs\GOLDEN_REGRESSION_FIXTURES.md
docs\GOLDEN_VIDEOS.md
```

18. assistant (2026-06-19T15:47:23.612Z)

Now let me carefully analyze RALLY_QUALITY_RESEARCH.md. This is a foundation doc dated 2026-06-16. I need to distinguish CURRENT-corpus claims from clearly-historical/dated ones. The key thing: this doc has §3.0 "Corpus today: 6 golden videos" (line 148) and §4.3 mentions "~6 ? ~16 videos". I need to be careful here ? the §7.2 progress table is clearly tracking historical PRs (E1/W1/W2 measured at n=6) which are legitimate historical records. But "Corpus today: 6 golden videos (label-set hash `50d98ffbc370`)" is a PRESENT-TENSE claim about the current corpus.

Let me focus. The task says: only fix claims about the CURRENT/present corpus size or ingest status; leave historical/measured-at-n notes alone. Let me check the GOLDEN_SET docs now and also re-examine which RALLY_QUALITY_RESEARCH lines are present-tense.

19. user (2026-06-19T15:47:27.012Z)

```
10:[Omitted long matching line]
152:| **Roora** (Roora ML Pvt Ltd) | **Bengaluru** ? HSR Layout (verified, `rooragrp.com`) |
Image/video/text annotation; sports industry listed. **Selected for the initial paid pilot
(2026-06-02)** ? existence/capability/location verified; **quality TBD via the pilot**. |
**Verified** (capability) . pilot pending |
182:[Omitted long matching line]
```

20. user (2026-06-19T15:47:28.922Z)

```
1:# Golden-Set Subproject ? Design & Implementation Overview
3:**Status:** In progress . **Date:** 2026-05-30
4:**Companion docs:** design rationale + vendor shortlist in
```

GOLDEN_SET_VENDORING.md (ADR-006); scoring mechanics in EVALUATION.md.

6: This is the **implementation tracker** for turning the golden-set/vendor **design** into shipped code. It's split into small, independently-reviewable PRs. The whole subproject is **vendor-agnostic by construction** ? only one slice (GS-3, `HumanVendorProvider`) depends on which vendor you pick offline; everything else can land first.

8: [Omitted long matching line]

21. user (2026-06-19T15:47:30.063Z)

1: # Phase-1 Golden-Set Video Collection Spec

4: **Companion docs:** GOLDEN_SET_VENDORING.md (vendor process, ADR-006) · VIDEO_RECORDING_GUIDELINES.md (capture spec the user-facing guide derives from) · scoring in EVALUATION.md.

6: What footage to gather for the **first round** of golden-set creation so the trained rally segmenter reaches good precision/recall ? and so the eval number actually predicts production. Golden-set creation is **iterative**: expect multiple rounds, expanding toward whatever axis round-1 metrics show is weakest.

14: Our product ingests **users' own match recordings** (a fixed phone/GoPro on a tripod). Therefore:

16: - The golden set ? **especially the held-out eval split** ? must be dominated by **static, self-recorded, full-court** footage that mirrors real users.

48: > These are **literature-backed starting points**; the published minimums get finalized empirically against our golden set (VIDEO_RECORDING_GUIDELINES.md §3).

78: > **Guardrails** ? TODO (design with the user): train and eval live in **separate catalogs** today, kept apart by convention. We want hard guardrails ? permissions / explicit approval / re-confirmation at ingest & export ? so a video can **never** silently land in both. Tracked in BACKLOG.md; to be designed in a later session.

22. assistant (2026-06-19T15:47:33.875Z)

The GOLDEN_SET_VENDORING.md, GOLDEN_SET_IMPLEMENTATION.md, and GOLDEN_SET_PHASE1_VIDEOS.md don't contain stale CURRENT corpus-size claims ? they're vendor-process/design docs that talk about the vendoring track abstractly. Let me verify the GOLDEN_SET_VENDORING line 10 (long matching line) to be sure.

23. user (2026-06-19T15:47:35.343Z)

```
1      # Golden-Set Generation & Vendor Process (v1)
2
3      Status: Design v1 · Date: 2026-05-30 · Vendor shortlist added 2026-05-30
($6)
4      Owner decisions captured: licensed/owned video only · rally `[start,end)` depth ·
tiered turnaround (fast automated + slow human) · vendor bake-off · Bangalore/India-local
vendor preference
5
6      This document defines how we produce golden-set ground truth (hand-verified rally
boundaries) for the badminton highlights product, how we select annotation vendors, and how
that same machinery powers on-demand / surprise regression testing of the production tool.
7
8      > Companion docs: the scoring mechanics live in [EVALUATION.md](EVALUATION.md); the
annotated-corpus rationale is [ADR-002](archives/decisions/DECISIONS.md); the decision record
for this process is ADR-006. Video-capture quality requirements (a dependency of
trustworthy golden sets) are tracked separately in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md). The regress() +
baseline.json machinery in §5 is generalized into a variant A/B + promotion gate in
[EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md).
9
10     > Update (2026-06-02): §6 shortlist done; the founder selected Roora (Bengaluru)
for the initial paid pilot ? a single-vendor pilot first, not the full bake-off below.
Roora's site/capability/location are verified; quality is TBD via the pilot. Step 0 (the
annotation guideline) is authored ? collector repo docs/annotation/` (ROORA_BRIEF.md rally
CSV schema + ANNOTATION_GUIDE.md illustrated guide + intake-manifest & NDA templates). Pilot
scope is 3 sports (badminton, pickleball, table tennis); the collector's import-local`
```

ingestion for all three is built. Still pending: the internal **gold key** and the vendor-dependent **GS-3 HumanVendorProvider** (GS-1/GS-2/GS-4 already landed ? see GOLDEN_SET_IMPLEMENTATION.md).

```

11
12    ---
13
14    ## 1. The keystone insight: one scorer, three jobs
15
16    A vendor's golden-set output is a list of rally `[start,end)` intervals ? structurally identical to a prediction from our pipeline. So the exact same deterministic scorer in `backend/eval/metrics.py` (`interval_iou` ? `match_intervals` ? `score`) grades all three of:
17
18    1. Product vs. ground truth ? the precision/recall/F1/IoU we already report (`EVALUATION.md`).
19    2. Vendor vs. our internal answer key ? the quality gate during vendor selection.
20    3. Annotator vs. annotator ? inter-annotator agreement (IAA), our proxy for "is this even a well-defined task?"
21
22    This symmetry is what makes the whole system small, auditable, and extensible. No new scoring code is needed ? only a new source of intervals (the vendor).
23
24    ---
25
26    ## 2. Why this also solves the privacy nuance
27
28    The monetization goal (hosted API) carries a "someone views the user's video" concern. This process sidesteps it by construction:
29
30    - Golden sets are built only from footage we license or own ? starting with the founder's constantly-growing personal collection of match videos, plus ShuttleSet (ADR-003) and any purpose-recorded clips. No end-user uploads ever flow to a vendor.

```

24. user (2026-06-19T15:47:37.774Z)

```

1:# Golden Regression Fixtures ? video-free, non-attributable regression suite
3:> Status: `IN PROGRESS` ? owner-gated . Owner: Avi . Created: 2026-06-16 . Last updated: 2026-06-16 . Legal: owner accepted the researched risk 2026-06-16 (conditions $3 stand; this is the owner's informed acceptance, not a retained-counsel opinion)
4:> Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE, per archive policy)
6:> Relation to existing docs: EXTENDS (does not supersede) `docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` / `docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
28:The rally-quality measurement is split across two machines by capability (`docs/GOLDEN_DATA_SHARING.md`):
29:- `backend/eval/fusion_golden.py` extracts `*_golden_features.json` ? needs the video + trajectory CSV + labels + GPU/WSL (GPU box only).
34:Goal: commit a tiny, version-tagged, numeric-only, non-attributable per-fixture corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression signal with zero video/GPU/DB, while never leaking attributable or sensitive data.
42:| D1 | Corpus is owner-recorded / owned | owner answer 2026-06-16 | Legal analysis takes Fork A (clean): no third-party copyright/ToS infringement; concern is trade-secret/moat + privacy. Strongest non-attributability (no public reference to correlate against). |
48:| D7 | Two tiers *(design)* | workflow-2 synthesis | Tier-0 public/no-key (shuttle-only, de-attributed). Tier-1 key-gated/out-of-repo (person/audio, full corpus), skips cleanly when absent*. |
58:| India (primary) | Not copyrightable as facts (*EBC v. D.B. Modak*); but compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ? only bites if lifted from a 3rd-party dataset, not self-derived | None (no sui-generis right) ? a real permissiveness edge | Anti-scraping ToS (untested in court) + IT Act s.43/s.66 civil/criminal exposure for 3rd-party ingestion | Owned-source + minors-excluded + no person/pose = clean |
79:Freeze at the trajectory CSV ? CONTROL-path replay. One shared helper `backend/eval/golden_fixtures.py::replay_fixture()` is the single code path used by both the freeze tool and both gates. As shipped, that path is (per the module docstring):

```

```

88:| **Tier-1** | OneDrive (default) **or** SOPS-encrypted in-repo | token/age key | full corpus
**with** person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | **Skips cleanly**
when key/data absent (generalize `_golden_present()` ? `_tier1_present()` ) |
113:[Omitted long matching line]
147:Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
152:[Omitted long matching line]
154:| **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
155:| **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macros) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |
156:| **M4** | **Quality-floor gate + first `regression_baseline.json`.
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
157:[Omitted long matching line]
161:| **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
162:| **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row to
`CODE_MAP $0` + a `golden_fixtures.py` entry to `CODE_MAP $2.3`; cross-linked
`GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186;
memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that
doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | `_this PR`
|
170:| [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced**; legal ? owner-accepted |
172:[Omitted long matching line]
181:3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-party
ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10** and
whether per-player tracking is prohibited s.9(3) monitoring of a minor.
209:[Omitted long matching line]
210:**Internal docs:** `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`,
`EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md
$0`.

```

25. user (2026-06-19T15:47:39.043Z)

```

1:# Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
4:**Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts (trajectory
CSVs, `_golden_features.json`, manifests, baseline snapshots) so the **GPU box** and **CPU
dev/agent sessions** share the same files ? without committing large data to git.
8:- **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the
**videos**, the **trajectory CSVs**, the **`.rallies.csv` golden labels**, and a **GPU/WSL**
stack ? only the GPU box can run it.
9:- **`backend/eval/fusion_compare.py`** / **`backend/eval/ablation.py`** (the LOVO **litmus**:
?F1, bootstrap CI, paired sign test) are **pure-CPU re-scoring** that read only the small
`_golden_features.json` ? any session (incl. a CPU dev container) can run them.
11:Today those artifacts sit in gitignored `output/` (per-machine) and scattered local folders
(`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs them
between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap with
OneDrive.
14:`%USERPROFILE%\OneDrive\rally-annotator\golden-data\` ? `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\`
19:golden-data/
21:  features/<corpus-tag>/          *_golden_features.json      (small ? the key shared artifact)
22:  trajectories/<corpus-tag>/      *_predict.csv / *_wasb.csv (medium ? reusable across runs)
23:  manifests/                      *_lovo_manifest.json
27:<corpus-tag>` = e.g. `2026-06-15-n6` (date + corpus size). **Do NOT** put raw videos here

```

(tens of GB) ? keep those local to the GPU box.

```

32: python -m backend.eval.fusion_golden --manifest <manifest> \
33:     --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"
35: (or extract to local `output/` then copy the `_golden_features.json` into the shared
folder).
39: python -m backend.eval.fusion_compare --features-dir "...golden-
data\features\2026-06-15-n6" --record
40: python -m backend.eval.ablation --features-dir "...golden-
data\features\2026-06-15-n6" --granularity group
42: ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/` to
also light up the `skipif` golden-litmus tests
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
45:- A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` /
`fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` /
`--features-dir` flags already make this a convention; the env var is just ergonomics.)
46:- A tiny `scripts/sync_golden.py` to copy `output/*_golden_features.json` ? the shared
folder.
50:- The features derive from the Proprietary corpus ? keep the OneDrive folder private
(no public share links).
52:[Omitted long matching line]

```

26. assistant (2026-06-19T15:47:48.332Z)

The GOLDEN_DATA_SHARING.md uses `2026-06-15-n6` as an EXAMPLE corpus tag (line 27 explicitly says "e.g." and "date + corpus size"), so that's illustrative, not a stale CURRENT-corpus claim. GOLDEN_REGRESSION_FIXTURES.md talks about "full corpus" abstractly without a stale n. These aren't in scope for staleness.

Now let me focus on the clear findings. The most authoritative present-tense claim is in GOLDEN_VIDEOS.md (the registry). Let me re-confirm the manifest's fps for the "ingestion in progress" entries to make sure GX030094 and Badminton_BXH_2 are now ingested (they are ? entries 10 and 11).

Let me also verify whether RALLY_DETECTION_GAP_CLOSURE.md's §3 line 40 "n=13" is a CURRENT claim, and look at the QUALITY_ITERATIONS re-grounding note more carefully to determine if "9 multicourt clips" is a present claim. Let me re-read the relevant context.

27. user (2026-06-19T15:47:49.315Z)

```

585     the proposal operating point** and re-run the #148 litmus to confirm the lift transfers.
The #148 litmus is
586     now the standing guard: it pins the honest measured behaviour on the present golden
corpus and trips if a
587     future windowing/corpus change silently shifts it.
588
589     > Re-grounded 2026-06-18. The original-6 single-court golden clips this litmus was
calibrated on have been
590     > retired from the box (their labels are gone); the present golden set is 9 multicourt
clips. On that
591     > recall-starved corpus (WASB tracks one shuttle while several courts rally ? served
over-seg ratio < 1) the
592     > served Gate-A ?F1 is now 0.034 (was 0.008), and under the permissive proposal
windowing the gate is
593     > -F1-neutral (?F1 ? 0.001) while still cutting over-seg (1.16?0.64) ? it no longer
reproduces the +0.074
594     > live. The canonical +0.074 / 6-of-6 / p=0.031 win remains a real result of the
original-6 corpus,
595     > preserved in the [first-A/B archive](archives/quality-iterations/2026-06-first-ab-
experiment/README.md).
596

```

28. assistant (2026-06-19T15:48:01.844Z)

This is a tricky one. The re-grounding note dated 2026-06-18 makes a present-tense claim "the

present golden set is **9 multicourt clips**". This is a measured-result note tied to a specific Gate-A re-measurement (?F1 ?0.034 measured on that 9-clip set). The ?F1 ?0.034 number was measured ON the n=9 set, so it's a historical measurement record. However, the phrase "the present golden set is 9 multicourt clips" is a CURRENT-corpus assertion that is now wrong (n=15).

The careful approach: the measured ?F1 ?0.034 was genuinely measured on n=9, so I shouldn't claim it's wrong. But "the present golden set is 9 multicourt clips" describing the CURRENT corpus contradicts n=15. Also "9 multicourt clips" is doubly wrong: the current corpus is 15, and it's not all-multicourt (the P7 table says 11 single-court + 4 multicourt). I'll make a surgical fix that corrects only the present-tense size claim while preserving the measured-at-n=9 result. Actually, the cleanest honest fix is to mark that the n=9 measurement was at the time of re-grounding. Let me phrase it to preserve the measurement's provenance.

Now let me check the git blame / recent commits context to confirm the n=9 re-grounding was a recent prior-session artifact, and verify line 62 already updated to n=15 (it was). Let me also double check the GAP_CLOSURE §3 line 40 and §6 are present claims. Let me re-read those precisely.

The findings I'm confident about. Let me now also scan QUALITY_ITERATIONS line 18 context once more and verify there isn't a more current corpus statement that supersedes it nearby. Line 18 is in the header/preamble (present-tense definitional). It says "Golden corpus = the 6 labelled matches" ? definitively a present-tense CURRENT claim, now wrong (n=15), and it even enumerates only the original 6 names.

Let me verify the exact unique text strings for each edit by checking surrounding uniqueness.

29. user (2026-06-19T15:48:04.015Z)

```
docs\GOLDEN_VIDEOS.md:28:## Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 9)
docs\GOLDEN_VIDEOS.md:42:## Labeled, ingestion in progress (WASB trajectory extracting)
docs\QUALITY_ITERATIONS.md:18:**Golden corpus = the 6 labelled matches** in
`output/human_lovo_manifest.json` (owner-confirmed
docs\QUALITY_ITERATIONS.md:109:**Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips
GX010137/GX010141) ? the milestone
docs\RALLY_QUALITY_RESEARCH.md:147:- **Corpus today: 6 golden videos** (label-set hash
`50d98ffbc370`); ~10 more expected via the
docs\RALLY_DETECTION_GAP_CLOSURE.md:12:[Omitted long matching line]
docs\RALLY_DETECTION_GAP_CLOSURE.md:26:| D3 | **Golden labels = ground truth; Gemini =
reference, not truth** | n=12 corpus | Gemini under-detects too ? agreement ? accuracy. Report
both, gate on golden. |
docs\RALLY_DETECTION_GAP_CLOSURE.md:64:- **n is small.** One golden clip scored so far
(GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as
the owner labels more (GX010137 / GX020140 / a large-count clip next).
docs\RALLY_DETECTION_GAP_CLOSURE.md:89:- [ ] Golden corpus covers the Boxhill set + enough
variety that per-video reads are trustworthy (n meaningfully > 12).
```

30. assistant (2026-06-19T15:48:19.000Z)

Now I have a comprehensive picture. Let me carefully classify each candidate for staleness, applying the rule "only fix CURRENT/present claims; leave historical/measured-at-n notes alone."

Let me analyze each:

```
**GOLDEN_VIDEOS.md (the registry ? the primary target):**
1. Line 28: `## Ingested golden corpus ? ... (n = 9)` ? **STALE**: this header is the registry's
CURRENT size claim; manifest has 15. HIGH severity (this is "the single source of truth").
2. Lines 42-47: the "Labeled, ingestion in progress" section lists GX030094 and Badminton BXH 2
as "WASB running ? ingest next" ? both are now ingested (manifest entries 10, 11). **STALE**:
their ingest status is wrong.
3. The registry table (lines 30-40) only lists 9 rows; missing 6 ingested clips (GX030094,
Badminton_BXH_2, GX010141, GX010137, Boxhill_Doubles, GX020094). This is an incompleteness, but
the load-bearing stale CLAIM is the "n = 9" header + the "ingestion in progress" mislabeling.
```

```
**QUALITY_ITERATIONS.md:**
```

4. Line 18: "Golden corpus = the 6 labelled matches" ? ****STALE present-tense definitional claim****. n=15. HIGH.

5. Line 590: "the present golden set is ****9 multicourt clips****" ? present-tense claim, now wrong (15, and not all multicourt). But the ?F1 ?0.034 was measured on that n=9 set. I'll surgically fix only the present-tense size assertion while preserving the measurement provenance.

6. Line 109 "Re-verified on the grown n=13 corpus" ? ****HISTORICAL**** (dated 2026-06-18 changelog of a re-verification that happened at n=13). Leave alone per instructions.

****RALLY_QUALITY_RESEARCH.md:****

7. Line 147-148: "Corpus today: 6 golden videos (label-set hash `50d98ffbc370`)" ? ****STALE present-tense claim**** ("Corpus today"). HIGH. The §7.2 table rows measured at n=6 are historical and I leave those.

****RALLY_DETECTION_GAP_CLOSURE.md:****

8. Line 12 (TL;DR): "the corpus is n=12" / "n=13 corpus" ? let me re-read line 12 exact text.

9. Line 26 (D3): "n=12 corpus" ? this is a source-citation for a decision; borderline. Let me examine.

10. Line 64 (§6): "the corpus is n=12 with several weak clips" ? present-tense **CURRENT** claim. **STALE**.

11. Line 89 (§9 DoD): "n meaningfully > 12" ? this is a **TARGET** ("covers ... enough variety"), a goal threshold, not a current-size claim. Leave it (it's a definition-of-done bar).

12. Line 40 (§3): "Corpus aggregate (n=13, milestone)" ? measured aggregate. Mixed. Let me check.

Let me re-read line 12 and the §3 line 40 context precisely to classify §3 numbers.

31. user (2026-06-19T15:48:19.879Z)

11 ## 0. TL;DR

12 We are using the ****growing golden corpus**** (human rally labels) to find and close the rally-detection accuracy gap ****per video, before launch**** ? and the first ground-truth clip already reshaped the picture. The boundary work (R2/R3/R4) ****holds**** ? when local fires on a real rally, its edges are tight ($|?| < 1$ s). ****The remaining gap is DETECTION, not boundaries:**** (G1) ****recall**** ? both local and Gemini miss ~half the rallies on hard clips; (G2) ****precision**** ? local over-fires (false positives), especially on continuous drilling; (G3) ****rally-END over-extension**** ? when a player **quickly knocks the shuttle back after a point**, shuttle-velocity alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the priority: ****LOCAL = high recall / low precision, GEMINI = high precision / low recall**** (opposite, complementary ? on the normal clip local's recall **beats** Gemini's). So ****the #1 lever is precision**** (cut local's false positives via ``rally_gate`` Gate-A + colocation; the player-state rally-END rule feeds this), with ****signal fusion**** (local's recall + Gemini-style precision) as the durable win. ****Caveat:**** two golden clips so far (n=13 corpus, GX010141+GX010137) ? directional, not conclusive.

32. assistant (2026-06-19T15:48:38.576Z)

Line 12 TL;DR ends with "two golden clips so far (n=13 corpus, GX010141+GX010137)". The "two golden clips so far ... GX010141+GX010137" is the n=2 ground-truth-scored finding (historical/measured), but the parenthetical "(n=13 corpus...)" states the **CURRENT** corpus size as 13 ? now wrong (15). This is a present-tense corpus-size assertion embedded in an otherwise-historical finding.

Now for §3 line 40: "Corpus aggregate (n=13, milestone): tIoU F1@0.5 0.474 ...". This is a measured aggregate AT n=13. That's a historical measurement (the milestone numbers were computed on n=13). Per the rule "leave clearly-HISTORICAL notes (at n=11 filed, changelog entries) alone ? only fix claims about the **CURRENT**/present corpus size", a measured aggregate at n=13 is analogous to "at n=11 filed" ? it's a result computed on a specific corpus snapshot. I'll leave line 40's measured number alone. Similarly line 64's "the corpus is n=12" is a present-tense statement of current size ? that one IS stale. And line 26's D3 "n=12 corpus" is a citation/source column.

Let me be principled and consistent:

- ****Present-tense statements of the CURRENT corpus size**** ? STALE, fix.
- ****Measured results computed at a specific n**** ("aggregate (n=13)", "Re-verified ... n=13", "measured n=15") ? historical snapshots, leave alone.

- **Targets/thresholds** ("n meaningfully > 12") ? leave.

Applying this:

- Line 12: "(n=13 corpus, GX010141+GX010137)" ? the n=13 here is describing the current corpus the two clips sit in. It's a present-corpus-size claim ? STALE.
- Line 26 (D3 source = "n=12 corpus"): this labels the decision's evidence basis; it's describing the corpus at decision-time, borderline-historical. But it reads as a present-corpus characterization. I'll treat it as a CURRENT claim ? low severity.
- Line 64: "the corpus is n=12 with several weak clips" ? explicit present-tense CURRENT size ? STALE, medium.

Let me check that §6 line 64 also says "One golden clip scored so far (GX010141)" ? but §3 and the changelog show GX010137 was ALSO scored (n=2). So line 64 "One golden clip scored so far" is itself stale vs the doc's own §3/changelog. However the task slice is corpus SIZE/ingest status; "one clip scored" is about ground-truth-scoring progress, related but the explicit size claim "the corpus is n=12" is the clean target. I'll fix the size and leave the "one clip scored" wording since §0/§3 already supersede it and it's not strictly a corpus-size/ingest claim. Actually, to be safe and surgical, I'll only touch the "n=12" size token.

Let me verify uniqueness of stale_text strings for the edits I'll propose, particularly the GAP_CLOSURE n=12 occurrences (there are 2: line 26 and line 64) and n=13 (line 12 and line 40).

The strings differ enough by surrounding context. Now let me confirm there are no OTHER stale size claims I'm missing across the docs slice (e.g., GOLDEN_VIDEOS table count, an n=11 present claim elsewhere).

33. user (2026-06-19T15:48:40.468Z)

12:[Omitted long matching line]

26:| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |

40:- **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25 0.633; R2 to F1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above the mean. `[verified]`

64:- **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).

34. assistant (2026-06-19T15:48:51.924Z)

I have all the data I need. Let me finalize my findings. For the GOLDEN_VIDEOS.md ingestion-in-progress section, the cleanest surgical fix is to update the n=9 header and the section header + status. But I must produce minimal surgical edits that don't require me to fabricate the full 15-row table (I can't invent video_id/rally-count details for the 6 missing rows). The load-bearing stale CLAIMS are:

- the "(n = 9)" count ? correct to "(n = 15)"
- the "ingestion in progress" status for GX030094 + Badminton BXH 2 (now ingested)

For the section header/status, the honest minimal fix is to note those two are now ingested. Since I cannot reconstruct full table rows with verified video_ids/rally counts, I'll fix the explicit stale tokens (the count and the status line) and flag the table incompleteness in the why. Let me make the status fix surgical: the section title "Labeled, ingestion in progress (WASB trajectory extracting)" and the per-row "WASB running ? ingest next" status are the stale claims.

Let me verify the exact fps of GX030094 (29.97) and Badminton_BXH_2 (59.94) match the doc's claims ? they do (line 46: 29.97; line 47: 59.94). Good, so these two ARE the now-ingested clips. I'll correct the n=9 header and mark these as ingested.

Now let me write up all findings.

35. user (2026-06-19T15:49:20.749Z)

Structured output provided successfully

36. assistant (2026-06-19T15:49:35.767Z)

I've completed the corpus-staleness audit. Verified ground truth:
`output/human\_lovo\_manifest.json` has **15 entries** (the 15 names listed in the task), of which
11 are single-court and 4 multicourt per the P7 stratified table.

Found 8 code-contradicting staleness items, all verified against the manifest:

docs/GOLDEN_VIDEOS.md (the registry ? primary target):

- Line 28: registry header says `(n = 9)` ? should be 15 (table also lists only 9 rows, missing 6 ingested clips). HIGH.
- Lines 42-47: GX030094 and Badminton BXH 2 listed as "ingestion in progress / WASB running ? ingest next" but both are already ingested (manifest entries 10 and 11, fps matches). HIGH.

docs/QUALITY_ITERATIONS.md:

- Line 18: "Golden corpus = the 6 labelled matches" ? present-tense definitional claim, now n=15. HIGH.
- Line 590: re-grounding note "the present golden set is **9 multicourt clips**" ? wrong size (15) and wrong composition (11 single + 4 multi). MEDIUM. Surgically preserved the genuinely-measured-at-n=9 ?F1 ?0.034 result by re-casting the size as the corpus state at re-grounding time.

docs/RALLY_QUALITY_RESEARCH.md:

- Line 147: "Corpus today: 6 golden videos" ? present-tense current-corpus claim, now 15. MEDIUM. (Left the \$7.2 historical n=6 measured rows intact.)

docs/RALLY_DETECTION_GAP_CLOSURE.md:

- Line 12 (TL;DR): "(n=13 corpus, ...)" ? 15. MEDIUM.
- Line 64 (\$6): "the corpus is n=12" ? 15. MEDIUM.
- Line 26 (D3 source): "n=12 corpus" ? 15. LOW.

Deliberately left alone as clearly-historical / not-in-scope (per the rule to fix only CURRENT corpus-size/ingest claims):

- QUALITY_ITERATIONS.md line 62 "n=15 golden" (already correct), line 109 "Re-verified on the grown n=13 corpus" (dated historical re-verification), and all the per-tier measured-at-n=6/n=8/n=11 entries.
- RALLY_DETECTION_GAP_CLOSURE.md line 40 "Corpus aggregate (n=13, milestone)" (a measured aggregate snapshot) and line 89 DoD "n meaningfully > 12" (a target threshold).
- GOLDEN_SET_VENDORING.md / GOLDEN_SET_IMPLEMENTATION.md / GOLDEN_SET_PHASE1_VIDEOS.md / GOLDEN_DATA_SHARING.md ("2026-06-15-n6" is an explicit *example* corpus tag) / GOLDEN_REGRESSION_FIXTURES.md ? no stale current-corpus-size claims.